

Mechanical Engineering Project - Furuta Pendulum System Control

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Chapter 1

Introduction and system modeling

1.1 Introduction

In this project, our aim was to control a system consisting of a DC motor that applies torque to a Furuta pendulum, which includes a shaft, arm, and pendulum, with the input being the motor voltage. To control it using LTI methods, a linear model is required, so both the mechanical system and the brushed DC motor will be modeled as one linear system. Since the Furuta Pendulum is non-linear, it must be linearized around an equilibrium point. The project involves modeling, linearization, parameter identification, and designing a control strategy for implementation. **All quantities are expressed in SI units.**

1.2 System modeling

1.2.1 The test system

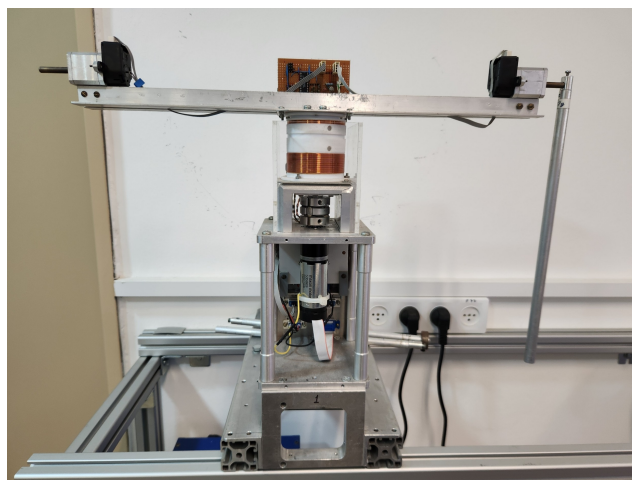


Figure 1.1: The Lab System

The system in Figure 1.1 is the Furuta pendulum as explained above. The only input of the system is the Voltage given to the motor.

The system has 2 encoders: The encoder near the motor for the motor's shaft angle, and an encoder near the pendulum for reading its rotation angle. While a mechanical system consisting of only the arm is linear from the voltage to its angle, the furuta pendulum that includes the pendulum is non-linear from voltage to the angles. In order to control the system using linear control tools, a linearized model will be used.

1.2.2 Defining the Mechanical System

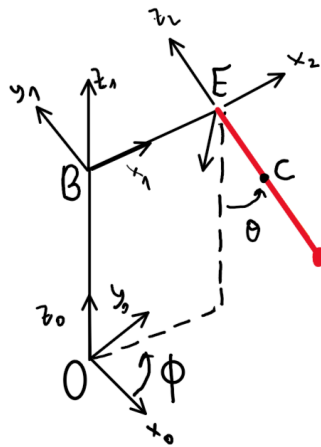


Figure 1.2: Model of the Mechanical System

The system in Figure 1.2 consists of a horizontal beam with a rigidly connected vertical beam, called the arm, which can rotate at an angle ϕ as one rigid body. A pendulum is attached to a free-rotation axis at the end of the arm, rotating with angle θ . A DC motor applies torque to rotate both the arm and the pendulum. The motor rotates the moment of inertia of the gearbox in addition to that of the arm and pendulum.

Parameters and Signals

The system has the following parameters:

- $r[m]$ - The distance from the axis BO to point E
- $l_c[m]$ - The distance from point E to the COM of the pendulum, point C
- $m_p[kg]$ - Mass of the pendulum.
- $J_a[kg \cdot m^2]$ - Moment of Inertia of the arm (rigidly connected vertical and horizontal beam) and gearbox, around BO axis.
- $J_p[kg \cdot m^2]$ - Moment of Inertia of the pendulum, around its COM with axis perpendicular to BE.

- $f_a, f_p [J/rad]$ - Damping coefficients in the rotation of the arm and pendulum, respectively.
- $g = 9.8 [m/s^2]$ - The standard acceleration of gravity/free-fall

The following presents the systems signals and the ones used while obtaining the systems equations with Euler-Lagrange method:

- $\phi(t)$ - Angle of rotation of the arm.
- $\theta(t)$ - Angle of rotation of the pendulum.
- $q(t) = \begin{pmatrix} q_1(t) \\ q_2(t) \end{pmatrix} = \begin{pmatrix} \phi(t) \\ \theta(t) \end{pmatrix}$ - Vector of the mechanical systems DOF.
- $T(t) = V_a(t) + V_p(t)$ $[J]$ - Total kinetic energy (KE) of the system, where $T_a(t)$ and $T_p(t)$ are the contributions of the arm and pendulum respectively.
- $V(t)[J]$ - Total potential energy (PE) of the system, where $V = V_a + V_p$ similarly to the KE.
- $L(t) = T(t) - V(t)$ - The Lagrangian of the system.
- $D(t)$ - The dissipation function of the system, expresses the contribution of damping forces in the energy equations.
- $Q(t) = \begin{pmatrix} Q_1(t) \\ Q_2(t) \end{pmatrix}$ - Vector of generalized forces on the system.

Axis-Systems

The coordinate systems are defined as follows: System (x_0, y_0, z_0) is static and located at the base of the shaft. System (x_1, y_1, z_1) is attached to the arm in point B, and rotates by an angle ϕ around $+\hat{z}_0$, and system (x_2, y_2, z_2) is attached to the pendulum at point E and rotates by θ around $-\hat{x}_1$, as shown in fig. 1.2.

With this, the inertia matrix of the pendulum is given by:

$$I_{(c,2)} = \begin{pmatrix} J_p & 0 & 0 \\ 0 & J_p & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Where the inertia is being calculated around the COM of the pendulum, in an axis-system with the same orientation as axis-system 2.

Obtaining non-linear Equations

To derive the equations of motion for the mechanical system, we use the Euler-Lagrange method for analyzing physical systems:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} + \frac{\partial D}{\partial \dot{q}_i} = Q_i, \quad i = \{1, 2\} \quad (1.1)$$

To acquire the 2 equations that solve for the angles of the arm and pendulum, for input $M(t)$ and IC, we need to find the KE and PE: $T(t)$ and $V(t)$.

After getting the energy's as a function of the system parameters and angles, we substitute them in equation 1.1 and get this 2 non-linear equations of motion:

$$\begin{cases} (1.2.1) & J_a \ddot{\phi} + (J_p + m_p l_c^2) (\sin(2\theta) \dot{\theta} \dot{\phi} + \sin^2(\theta) \ddot{\theta}) \\ & + m_p (r^2 \ddot{\phi} - r \ddot{\theta} l_c \cos(\theta) + r \dot{\theta}^2 l_c \sin(\theta)) \\ & + f_a \dot{\phi} = M(t) \\ (1.2.2) & (J_p + m_p l_c^2) \ddot{\theta} - m_p r \ddot{\phi} l_c \cos(\theta) \\ & - \frac{1}{2} (J_p + m_p l_c^2) \sin(2\theta) \dot{\phi}^2 \\ & + m_p g l_c \sin(\theta) + f_p \dot{\theta} = 0 \end{cases} \quad (1.2)$$

Description in State Space and Equilibrium Points

We start the linearization by defining the following state vector for the system:

$$x = \begin{pmatrix} q \\ \dot{q} \end{pmatrix} = \begin{pmatrix} \phi \\ \theta \\ \dot{\phi} \\ \dot{\theta} \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

Using x we can get the equations of the system in a state-space representation.

Firstly, we can rewrite the equations in 1.2 like:

$$\begin{pmatrix} J_a + (J_p + m_p l_c^2) \sin^2(\theta) + m_p r^2 & -m_p r l_c \cos(\theta) \\ -m_p r l_c \cos(\theta) & J_p + m_p l_c^2 \end{pmatrix} \begin{pmatrix} \ddot{\phi} \\ \ddot{\theta} \end{pmatrix} = \begin{pmatrix} g_1 \\ g_2 \end{pmatrix}$$

Where:

$$g_1 = g_1(x, M) = M - (J_p + m_p l_c^2) \sin(2\theta) \dot{\theta} \dot{\phi} - m_p r \dot{\theta}^2 l_c \sin(\theta) - f_a \dot{\phi}$$

and:

$$g_2 = g_2(x) = \frac{1}{2} (J_p + m_p l_c^2) \sin(2\theta) \dot{\phi}^2 - m_p g l_c \sin(\theta) - f_p \dot{\theta}$$

This representation of the equations can be written as:

$$M_f \dot{x} = N_f$$

Where:

$$M_f = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & J_a + (J_p + m_p l_c^2) \sin^2(\theta) + m_p r^2 & -m_p r l_c \cos(\theta) \\ 0 & 0 & -m_p r l_c \cos(\theta) & J_p + m_p l_c^2 \end{pmatrix}$$

And:

$$N_f = \begin{pmatrix} x_3 \\ x_4 \\ g_1 \\ g_2 \end{pmatrix}$$

Thus $\dot{x} = M_f^{-1}N_f$ and the representation of the system in state-space:

$$\begin{cases} \dot{x} = f(x, M(t)) = M_f^{-1}N_f \\ y(t) = h(x, M(t)) = Cx = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} \phi \\ \theta \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \end{cases} \quad (1.3)$$

Now, to get the equilibrium points, we need to find the points (x_{eq}, M_{eq}) s.t.:

$$\dot{x}_{(x,M)=(x_{eq},M_{eq})} = f(x_{eq}, M_{eq}) = 0$$

And after solving the linear equations we get:

$$\begin{pmatrix} x \\ M \end{pmatrix}_{eq} = \begin{pmatrix} \phi \\ \theta \\ \dot{\phi} \\ \dot{\theta} \end{pmatrix}_{eq} = \begin{pmatrix} \phi_0 \\ \pi k \\ 0 \\ 0 \end{pmatrix}, \quad \forall \phi_0 \in \mathbb{R}, k \in \mathbb{Z}$$

In the physical system, this corresponds to 2 unique equilibrium points:

- Bottom eq. where ϕ is arbitrary and $\theta = 0$, meaning the pendulum is pointing downwards.
- Top eq. where ϕ is arbitrary and $\theta = \pm\pi$, meaning the pendulum is pointing upwards.

Bottom linearization of the system

Now that we have the non-linear equations and the eq. points, we can get the linear system from the torque $M(t)$ to the angles with deviation from the bottom eq. :

$$\dot{x} = Ax + BM, \quad x_\delta = x - x_{eq} = x$$

Where:

$$A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{m_p^2 g r l_c^2}{\alpha} & -\frac{f_a(J_p + m_p l_c^2)}{\alpha} & -\frac{f_p m_p r l_c}{\alpha} \\ 0 & -\frac{m_p g l_c (J_a + m_p r^2)}{\alpha} & -\frac{f_a m_p r l_c}{\alpha} & -\frac{f_p (J_a + m_p r^2)}{\alpha} \end{pmatrix}$$

And:

$$B = \begin{pmatrix} 0 \\ 0 \\ \frac{J_p + m_p l_c^2}{\alpha} \\ \frac{m_p r l_c}{\alpha} \end{pmatrix}$$

Where $\alpha \equiv (J_a + m_p r^2)(J_p + m_p l_c^2) - m_p^2 r^2 l_c^2$.

After solving this state equation, we can get the angles directly from the components of the state vector.

In addition, it can be shown that this LTI system is equivalent to the following 2 Linear ODEs:

$$\begin{cases} (J_a + m_p r^2)\ddot{\phi} - m_p r l_c \ddot{\theta} + f_a \dot{\phi} = M \\ -m_p r l_c \ddot{\phi} + (J_p + m_p l_c^2)\ddot{\theta} + f_p \dot{\theta} + m_p g l_c \theta = 0 \end{cases} \quad (1.4)$$

Top linearization of the system

Similarly to the bottom equilibrium, we can get the upper equilibrium linear system with deviation from it:

$$\dot{x}_\delta = A_t x_\delta + B_t M, \quad x_\delta = x - x_{eq,top}$$

Where $x_{eq,top} = (0, \pi, 0, 0, 0)^T$ and:

$$A_t = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{m_p^2 g r l_c^2}{\alpha} & -\frac{f_a(J_p + m_p l_c^2)}{\alpha} & \frac{f_p m_p r l_c}{\alpha} \\ 0 & \frac{m_p g l_c (J_a + m_p r^2)}{\alpha} & \frac{f_a m_p r l_c}{\alpha} & -\frac{f_p (J_a + m_p r^2)}{\alpha} \end{pmatrix}$$

And:

$$B_t = \begin{pmatrix} 0 \\ 0 \\ \frac{J_p + m_p l_c^2}{\alpha} \\ -\frac{m_p r l_c}{\alpha} \end{pmatrix}$$

And like in the bottom eq., it can be shown that the equivalent 2 Linear ODEs are (with $\theta_\delta = \theta - \pi$):

$$\begin{cases} (J_a + m_p r^2)\ddot{\phi} + m_p r l_c \ddot{\theta}_\delta + f_a \dot{\phi} = M \\ m_p r l_c \ddot{\phi} + (J_p + m_p l_c^2)\ddot{\theta}_\delta + f_p \dot{\theta}_\delta - m_p g l_c \theta_\delta = 0 \end{cases} \quad (1.5)$$

1.2.3 Defining the DC-motor

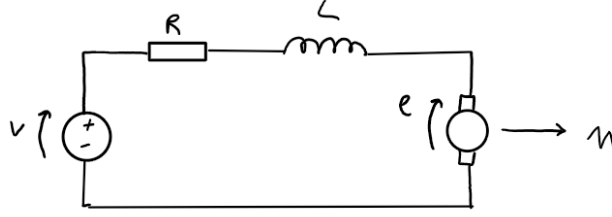


Figure 1.3: Model of DC motor

Figure 1.3 shows the model of the actuator used in the project - a Brushed DC motor. The model is of a linear electrical circuit that has input voltage $V(t)$, Resistance and Inductance of the motor R, L respectively and the output torque and current are $M_m(t), I(t)$ respectively.

The parameters:

- $R[\Omega]$ - Resistance of the rotor and the stator.
- $L[H]$ - Inductance of wires in the stator. Will be neglected in the DC motor's modeled dynamics.
- N_g - Gear-ratio of gearbox.
- $K_e[\frac{V \cdot s}{rad}]$ - Motor back EMF constant.
- $K_t[\frac{N \cdot m}{A}]$ - Motor torque constant.

Where the applied voltage is $V(t)[V]$, the motor current $I(t)[A]$ and the torque and angle output of the motor are $M_m(t)[Nm]$, $\phi_m(t)$, respectively.

The equations of the DC motor are known and they are, after neglecting the inductance L of the motor:

$$\begin{cases} RI(t) = V(t) - K\dot{\phi}_m(t) \\ M_m(t) = KI(t) \\ \frac{M(t)}{M_m(t)} = \frac{\phi_m(t)}{\phi(t)} = N_g \end{cases} \quad (1.6)$$

1.3 Block Diagrams of the System

1.3.1 Block Diagram of the whole system, under bottom eq.

Now that we have the ODEs of the motor and the linearized mechanical system, we can get the block diagram from the input voltage $V(t)$ to the output angles $\phi(t)$ and $\theta(t)$.

To obtain the block diagram of the system, let us first rearrange equations 1.4 and 1.6:

Firstly, we use the Laplace Transform on the second equation in 1.4 and get:

$$\frac{\theta(s)}{\phi(s)} = \frac{m_p r l_c s^2}{(J_p + m_p l_c^2)s^2 + f_p s + m_p g l_c} := P_{\theta\phi}(s) \quad (1.7)$$

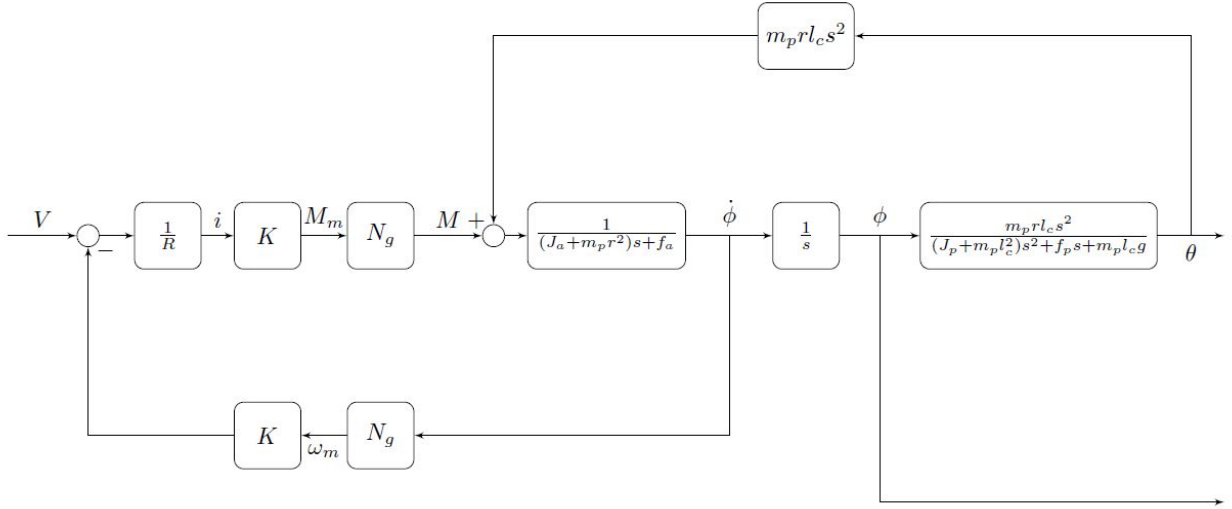


Figure 1.4: Block Diagram of the system under bottom equilibrium

And after using the Laplace Transform on the first equation, we can get:

$$s\phi(s) = \frac{1}{(J_a + m_p r^2)s + f_a} (M(s) + m_p r l_c s^2 \theta(s)) \quad (1.8)$$

Before drawing the block diagram, we also need the equations from the motor. From the voltage equality in 1.6, we get:

$$I(s) = \frac{V(s) - K s \phi_m(s)}{R} \quad (1.9)$$

And from the torque equation, while also using the gear ration, we get:

$$M(s) = N_g M_m(s) = N_g K I(s) \quad (1.10)$$

From these four equations (1.7,1.8,1.9,1.10), we can derive the block diagram of the system, which describes the relation between the voltage input $V(t)$ and the angles $\phi(t)$, $\theta(t)$. The resulting block diagram is shown in Figure 1.4. Using algebraic manipulation in the s -domain, we can derive the process from the voltage input to the arm angle:

$$P_{\phi V}(s) = \frac{N_1(s)}{D_1(s)} \quad (1.11)$$

Where:

$$N_1(s) = \frac{K N_g}{R} ((J_p + m_p l_c^2)s^2 + f_p s + m_p l_c g) \quad (1.12)$$

And:

$$\begin{aligned} D_1(s) = & ((J_a + m_p r^2)(J_p + m_p l_c^2) - (m_p r l_c)^2) s^4 \\ & + \left((J_a + m_p r^2)f_p + (J_p + m_p l_c^2)f_a + \frac{(K N_g)^2}{R} (J_p + m_p l_c^2) \right) s^3 \\ & + \left((J_a + m_p r^2)m_p l_c g + f_a f_p + \frac{(K N_g)^2}{R} f_p \right) s^2 \\ & + \left(f_a m_p l_c g + \frac{(K N_g)^2}{R} m_p l_c g \right) s \end{aligned} \quad (1.13)$$

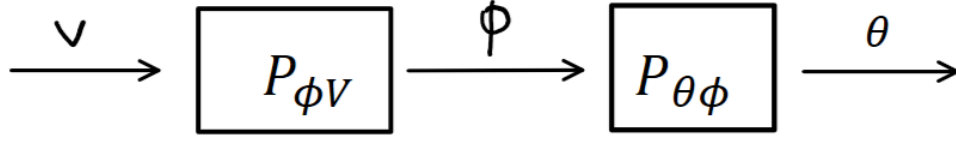


Figure 1.5: Simplified Block Diagram of the System

Notice the integrator in the system.

Now we can simplify the block diagram for our control purposes. The simplified Block Diagram of the system is shown in figure 1.5.

1.3.2 Block Diagram of the whole system, under top eq.

In a similar vain, we will obtain the block diagram and TF of the system in top equilibrium. Firstly, we use the Laplace Transform on the second eq. in 1.5 to get:

$$\frac{\theta_{\delta}(s)}{\phi(s)} = \frac{-m_p r l_c s^2}{(J_p + m_p l_c^2)s^2 + f_p s - m_p g l_c} := P_{\theta_{\delta}\phi}(s) \quad (1.14)$$

And after using the Laplace Transform on the first equation, we can get:

$$s\phi(s) = \frac{1}{(J_a + m_p r^2)s + f_a} (M(s) - m_p r l_c s^2 \theta_{\delta}(s)) \quad (1.15)$$

We will also use the motor equations in the same way we derived the bottom eq. block diagram. From these four equations (1.14,1.15,1.9,1.10), we can derive the block diagram of the system. The resulting block diagram is shown in Figure 1.6. Than, we can derive the process from the voltage input to the arm angle:

$$P_{\phi V}^t(s) = \frac{N_2(s)}{D_2(s)} \quad (1.16)$$

where:

$$N_2(s) = \frac{KN_g}{R} ((J_p + m_p l_c^2)s^2 + f_p s - m_p l_c g) \quad (1.17)$$

$$\begin{aligned} D_2(s) = & ((J_a + m_p r^2)(J_p + m_p l_c^2) - (m_p r l_c)^2) s^4 \\ & + \left((J_a + m_p r^2)f_p + (J_p + m_p l_c^2)f_a + \frac{(KN_g)^2}{R} (J_p + m_p l_c^2) \right) s^3 \\ & + \left(f_a f_p - (J_a + m_p r^2)m_p l_c g + \frac{(KN_g)^2}{R} f_p \right) s^2 \\ & + \left(-f_a m_p l_c g - \frac{(KN_g)^2}{R} m_p l_c g \right) s \end{aligned} \quad (1.18)$$

Notice the integrator in the system.

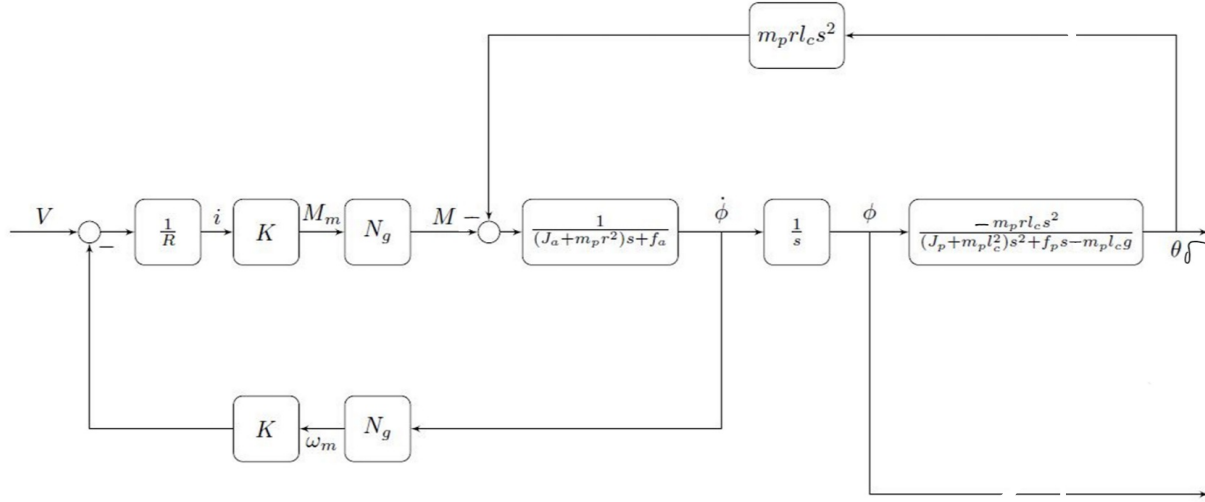


Figure 1.6: Block Diagram of the system under top equilibrium

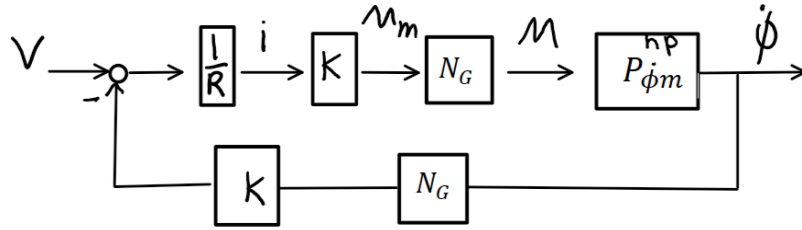


Figure 1.7: Block Diagram of the System without pendulum

1.3.3 Block Diagram of the system with no pendulum

In addition, if we remove the pendulum from the system, the equations of the mechanical system change into one simple equation:

$$J_a \ddot{\phi}(t) + f_a \dot{\phi}(t) = M(t) \xrightarrow{\mathcal{L}} s\phi(s) = \frac{1}{J_a s + f_a} M(s) \quad (1.19)$$

Notice that the system without the pendulum is linear, so no need for linearization.

From this equation and the motor equations, we can get the simplified block-diagram shown in figure 1.7. Where $P_{\dot{\phi}M}^{np}(s) = \frac{1}{J_a s + f_a}$ is the process from $M(t)$ to $\dot{\phi}(t)$, when there is no pendulum. The symbol "np" will be used to indicate process in the no-pendulum case. In this case, the process from the voltage to the arm angle:

$$P_{\phi V}^{np}(s) = \frac{\frac{1}{R} K N_g}{s(J_a s + (f_a + \frac{1}{R}(K N_g)^2))} \quad (1.20)$$

Now that we have the transfer functions of the system, we can go to the system identification step.

System Identification

We implemented the control scheme in Simulink and compiled the code using the dSPACE interface.

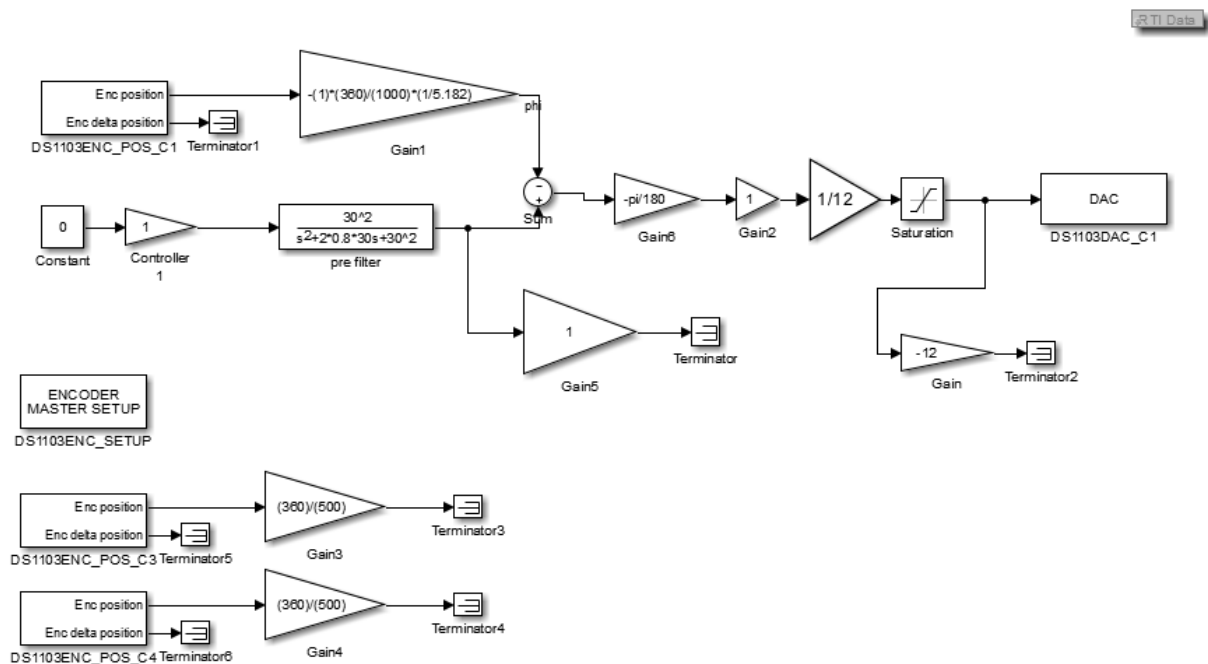


Figure 2.1: The First Simulink File

In the Lab, we designed the control scheme using a Simulink file on Matlab as in figure 2.1, which shows a control scheme of negative-unity feedback on $\phi(t)$ with $C(s) = 1$ used on the system (named Gain2). We used blocks that represent the readings from the encoders of the arm angle $\phi(t)$ and the pendulum angle $\theta(t)$.

The Simulink has a saturation block that bounds the voltage amplitude:

$$-12[V] < V(t) < 12[V]$$

From rotating the arm manually and seeing the readings on the encoders, we got that $N_g = 5.182$. We used this gear-ratio on the Simulink file to get $\phi(t)$ instead of $\phi_m(t)$.

2.1 Transfer Functions and Process of Identification

For our control, we found 3 transfer functions:

1. $P_{\phi V}(s)$ - The TF from $V(t)$ to $\phi(t)$ for system with pendulum, under bottom equilibrium
2. $P_{\theta\phi}(s)$ - The TF from $\phi(t)$ to $\theta(t)$ for system with pendulum, under bottom equilibrium
3. $P_{\phi V}^{np}(s)$ - The TF from $V(t)$ to $\phi(t)$ for system **without** pendulum, under bottom equilibrium

From the system's model, we saw that both TF's from $V(t)$ to $\phi(t)$ are not stable because of the integrator, and applying step input will make the angle diverge. Thus, we will use a simple control configuration of negative unity feedback that can stabilize both systems. In both cases, we used this control scheme with a proportional controller k_p .

Then, from the known input and measured output, we got the TF from the reference $r_\phi(t)$ to $\phi(t)$: $T_\phi(s)$, and then we can extract the plant with this, where $L_\phi(s) = P_{\phi V}(s)k_p$:

$$T_\phi(s) = \frac{L_\phi(s)}{1 + L_\phi(s)} \Rightarrow P_{\phi V}(s) = \frac{T_\phi(s)}{1 - T_\phi(s)} \frac{1}{k_p}$$

In contrast, we got $P_{\theta\phi}(s)$ directly from the measurements.

2.1.1 Identification of system without pendulum

Firstly, we closed a negative unity feedback loop on the system without the pendulum. We applied a step reference that was filtered using:

$$F(s) = \frac{30^2}{s^2 + 2 \cdot 0.8 \cdot 30s + 30^2} \quad (2.1)$$

Using the system identification toolbox on Matlab, we got that:

$$\widetilde{P_{\phi V}^{np}}(s) = \frac{2.978}{s(s + 1.316)} \quad (2.2)$$

The figure in 2.2 compares the simulation's response to the measured system's behavior with the same reference input. The simulation output resembles that of a 2nd-order undamped system. But in the real system, coulomb friction, which is non-linear and not modeled in the system's dynamics, makes the arm stop.

2.1.2 Identification of system with pendulum

After closing a loop on the system from $V(t)$ to $\phi(t)$. We got the following TF's:

$$(\text{Didn't use this TF}) \widetilde{P_{\phi V}}(s) = \frac{2.568s^2 + 0.5587s + 111.2}{s(s^3 + 2.269s^2 + 49.12s + 83.56)}, \quad \widetilde{P_{\theta\phi}}(s) = \frac{1.261s^2}{s^2 + 0.115s + 45.3} \quad (2.3)$$

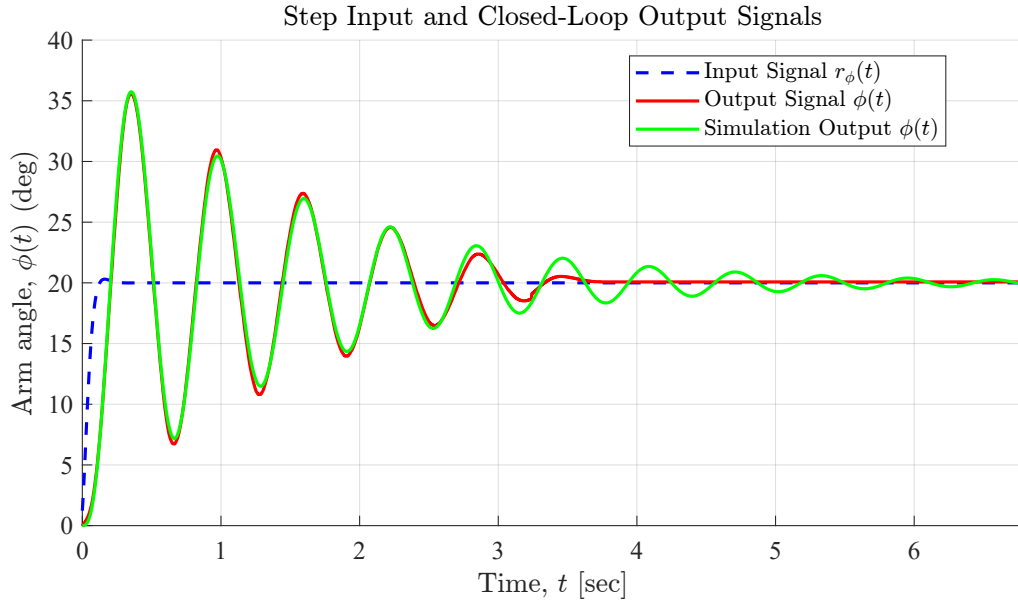
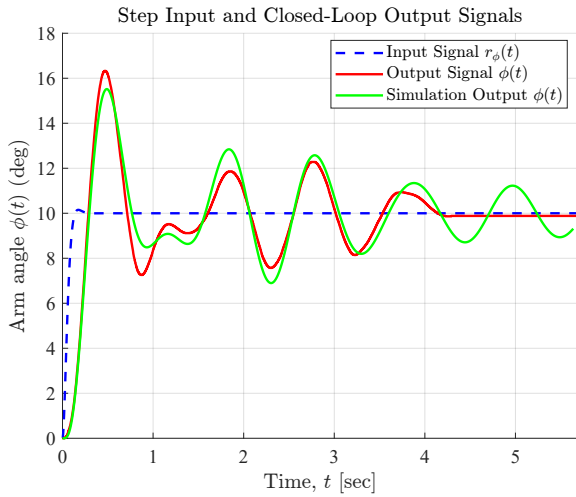
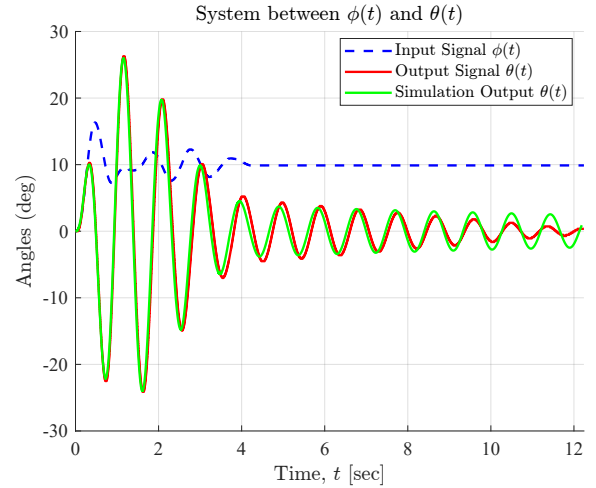


Figure 2.2: Shows $\phi(t)$ output from Lab and Simulation (with identified TF), for system without pendulum.



(a) $\phi(t)$ output from Lab and Simulation with identified TF



(b) $\theta(t)$ output from Lab and Simulation with identified TF

Figure 2.3: Comparison of system identification performance, for system with pendulum

In figure 2.3 we can see the comparison between the output of the lab and SI between $r_\phi(t) \rightarrow \phi(t)$ and between $\phi(t) \rightarrow \theta(t)$. After that experiment, we tuned the coefficients and got this TF:

$$\tilde{P}_{\phi V}(s) = \frac{3.08s^2 + 1.219s + 137.1}{s(s^3 + 1.451s^2 + 49.01s + 50.72)} \quad (2.4)$$

In conclusion, the TF's we will use for making our first control schemes will be: that in equation 2.2, the **second** on eq. 2.3 and from eq. 2.4.

Chapter 3

Tracking of the arm angle

The Tracking control scheme is shown in figure 3.1.

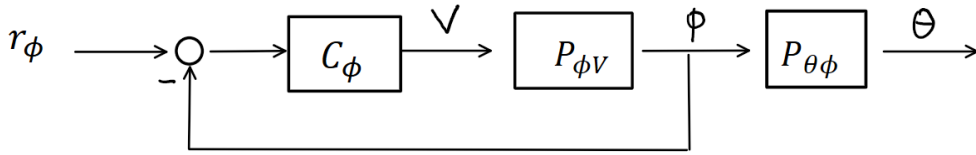


Figure 3.1: Unity-Feedback Control Scheme For Tracking of $\phi(t)$

In this chapter, we modified the scheme step-by-step in order to improve the tracking of $\phi(t)$, while reducing the required control input $V(t)$. We required the following from the closed-loop:

1. closed-loop stability
2. zero steady-state error for static reference $r_\phi(t) = 1(t)$
3. zero steady-state error for static disturbance $d(t) = 1(t)$
4. Adequate stability margins
5. The crossover frequency ω_c (in $[\frac{rad}{s}]$ from now on) is treated as a tuning parameter

In the Lab itself we used a step reference filtered through $F(s)$ as in equation 2.1.

3.1 Tracking Controller

3.1.1 Design of Tracking Controller using loop-shaping (in short - 'Loosh')

Firstly, we designed the tracking controller $C_\phi(s)$ for the control scheme in figure 3.1.

The original open-loop has $\omega_c = 1.52$. We designed a controller that has the following components: a gain setting $\omega_c = 25$, a lag controller with an integrator and a lead controller to set the PM. The resulting controller:

$$C_\phi(s) = \frac{349.3s^2 + 5217s + 2521}{s(s + 43.3)} \quad (3.1)$$

Closed loop stability is ensured through the Nichols diagram, zero steady-state error for static reference is obtained automatically from the integrator in $P_{\phi V}(s)$ and we added an integrator in $C_\phi(s)$ to reject static disturbances.

The Open-Loop of the Process $P_{\phi V}(s)$ itself with $C(s) = 1$ and with our designed $C_\phi(s)$ is shown in figure 3.2.

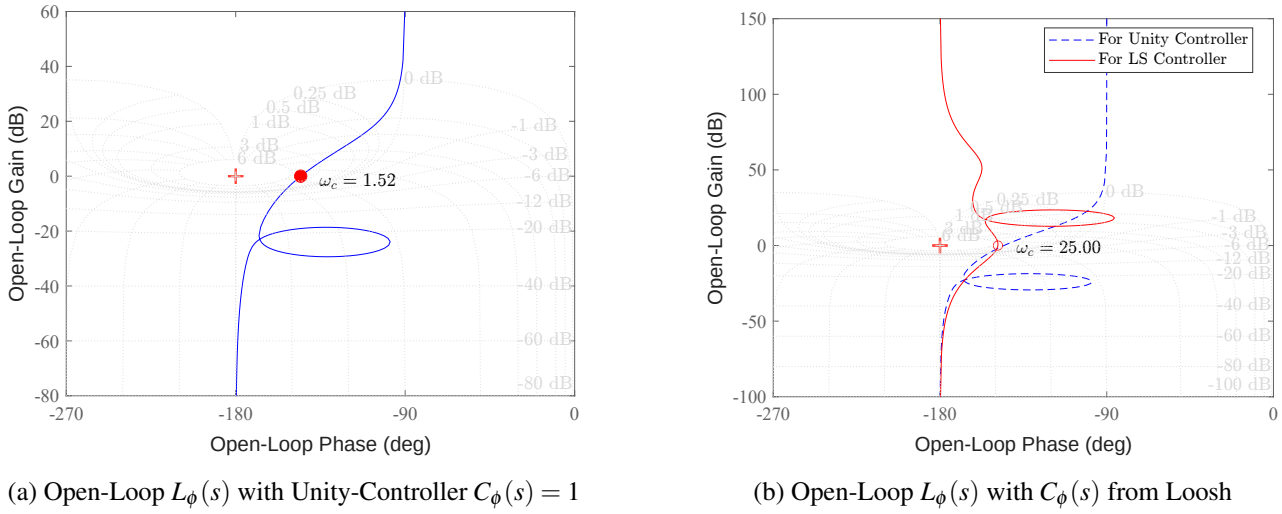


Figure 3.2: Comparison of Open-Loop with/without Loop-Shaping Controller

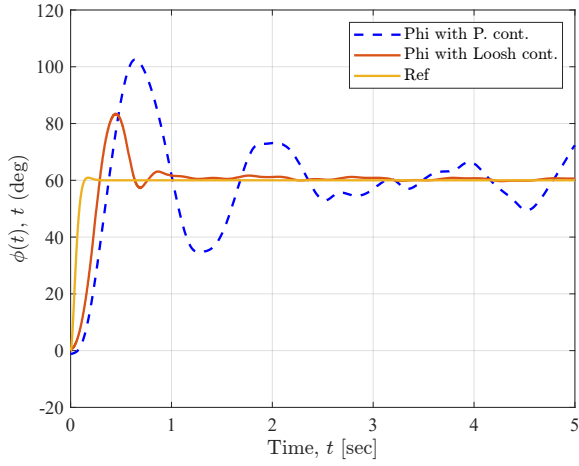
3.1.2 Comparing proportional with loop-shaping controller

We can see that in figure 3.3a that the second controller provides better tracking, with smaller OS and a shorter convergence time. Even so, from 3.3b we see that the required control effort while using this controller is reaching saturation, unlike with the gain controller.

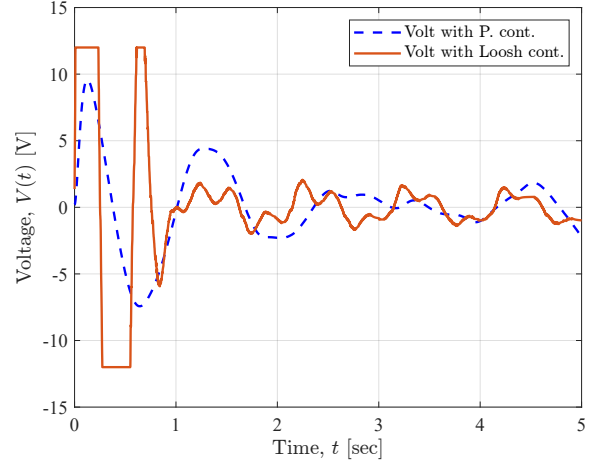
Saturation is undesired due to it being non-linear and deviating from the designed linear-control behavior. Thus, we would like **a reference that is less demanding** of our closed-loop system.

3.2 Time-Optimal Control Based Reference Signal

In eq. 1.11 we saw that the process $P_{\phi V}(s)$ is strictly proper and thus represents an inertial system. Hence, instead of using a step reference that is physically unattainable we formulated a reference that is based on the maximum control value that can be applied and the systems dynamics.



(a) Comparison of tracking controllers: $C_\phi(s) = 10$ and $C_\phi(s)$ from Loosh



(b) Comparison of Voltage: $C_\phi(s) = 10$ and $C_\phi(s)$ from Loosh

Figure 3.3: Comparison of Closed-Loop performance with/without Loop-Shaping Controller

3.2.1 Calculating the Bang-Bang reference Signal

Firstly, we formulated the problem. Consider the system from the Voltage to the Arm angle:

$$P_{\phi V}(s) = \frac{\phi(s)}{V(s)}$$

We would like to find the voltage $V(t)$ that would take the angle $\phi(t)$ from ϕ_0 to ϕ_f under minimum time, assuming initial rest, for the condition of $|V(t)| \leq V_{max}$. From Time-Optimal Control theory, we have these results for optimal $V(t)$:

1. Its non-zero for $0 < t < t_f$
2. It only applies maximum voltages $\pm V_{max}$
3. The number of switching required, for systems with only real poles, is $n - 1$, where n is the systems order.

To simplify the problem, we designed the optimal voltage with the assumption that the system behaves similarly to $P_{\phi V}^{np}(s)$ in eq. 1.20. In this case, the system used is in 2.2:

$$P_{\phi V}^{Virtual}(s) \approx P_{\phi V}^{np}(s) = \frac{\frac{1}{R}KN_g}{s(J_a s + (f_a + \frac{1}{R}(KN_g)^2))} = \frac{2.978}{s(s + 1.316)} = \frac{k}{s(\tau s + 1)}$$

Where: $\tau = 0.76$, $k = 2.263$. Meaning, the optimal voltage signal is bang-bang:

$$V_{opt}(t) = \begin{cases} V_1 & 0 < t < t_{sw} \\ -V_1 & t_{sw} < t < t_f \\ 0 & t_f < t \end{cases}$$

Where t_{sw} is the switching time between 0 and t_f , and V_1 is the initial voltage with $|V_1| = V_{max}$ but with unknown sign. We got the following formulas for the switching times and initial voltage sign:

$$\begin{cases} \text{sign}(V_1) = \text{sign}(\phi_f - \phi_0) \\ t_{sw} = \frac{|\phi_f - \phi_0|}{kV_{max}} + \tau \ln \left(1 + \sqrt{1 - e^{\frac{-|\phi_f - \phi_0|}{\tau kV_{max}}}} \right) \\ t_f = \frac{|\phi_f - \phi_0|}{kV_{max}} + 2\tau \ln \left(1 + \sqrt{1 - e^{\frac{-|\phi_f - \phi_0|}{\tau kV_{max}}}} \right) \end{cases} \quad (3.2)$$

Thus, the time-Optimal reference is:

$$r_\phi(t) = P_{\phi V}^{Virtual}(V_{opt}(t))$$

To make the voltage signal smoother, we filtered it through: $F_{BB}(s) = \frac{\omega_0}{s + \omega_0}$, where $\omega_0 = 100$.

3.2.2 Comparison of tracking references by Loop-Shaping controller

Firstly, we implemented the control law as seen in Figure 3.4.

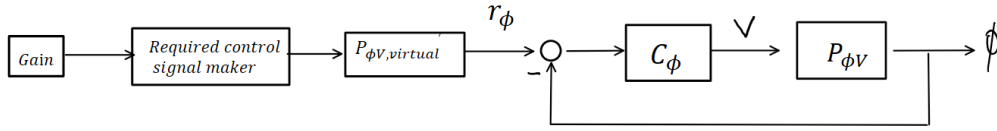
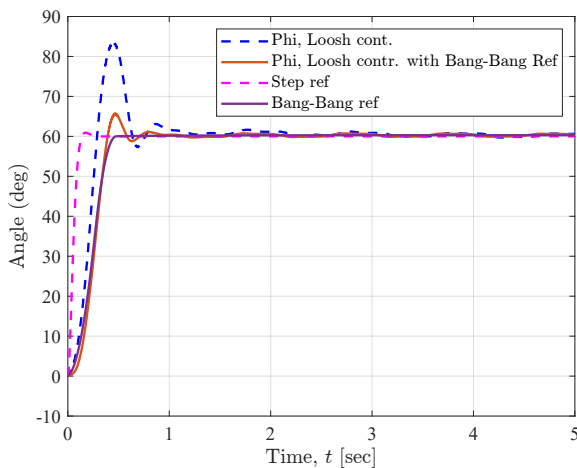
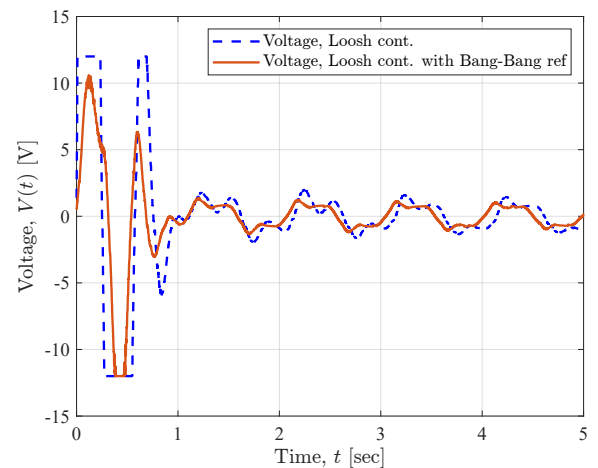


Figure 3.4: Tracking Closed-Loop with time-optimal reference, for 1DOF scheme.

In figure 3.5 we see that the time-optimal reference required less voltage, for the same ϕ_f, ϕ_0 and controller, and the tracking is better with less OS and a shorter convergence time.



(a) Shows Loosh controller tracking $r_\phi(t) = 60^\circ 1(t)$ and Bang-Bang reference, in the lab



(b) Shows $V(t)$ in both cases of the reference signal, in the lab

Figure 3.5: Comparison of Tracking and Voltage of Loosh controller, for step and bang-bang reference.

In figure 3.6, the actual applied voltage isn't similar to the desired bang-bang one, it doesn't respect the boundary's of voltage and contradicted our desire to bound the voltage below saturation.

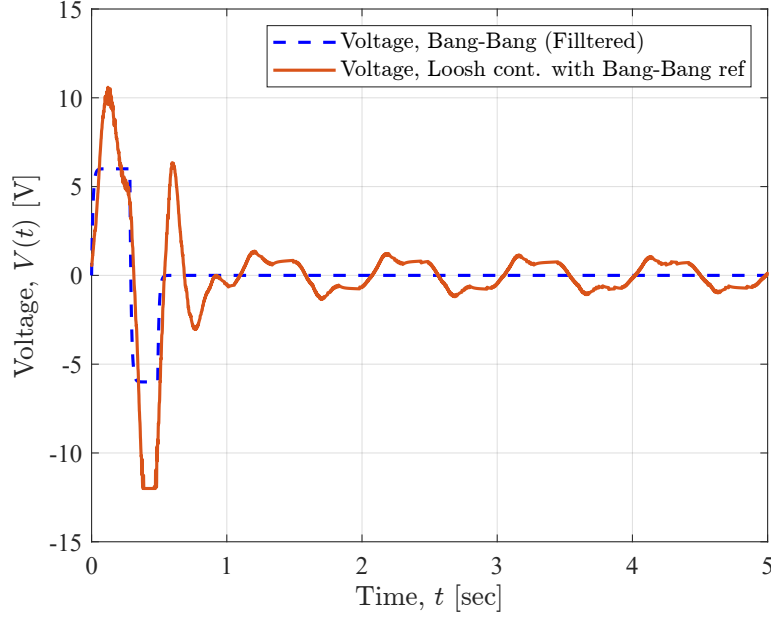


Figure 3.6: Comparing $V(t)$ applied in the lab under optimal reference w. $\phi_f = 60^\circ$ and the optimal voltage.

Consequently, we modified the control scheme to improve tracking.

3.3 Applying the 2DOF Control Scheme on the tracking Closed-Loop

3.3.1 The Scheme of a 2DOF Controller

To improve tracking, we modified the control scheme to include a 2DOF structure, as shown in figure 3.7.

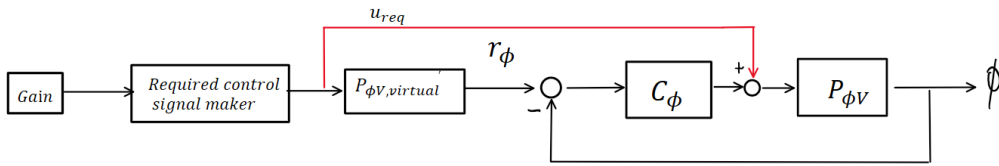
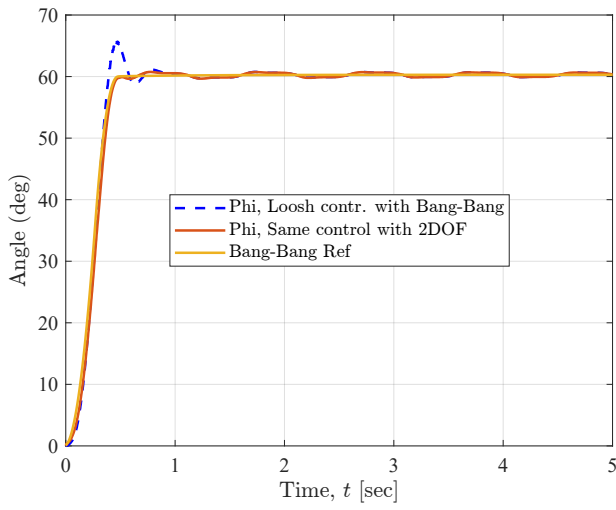
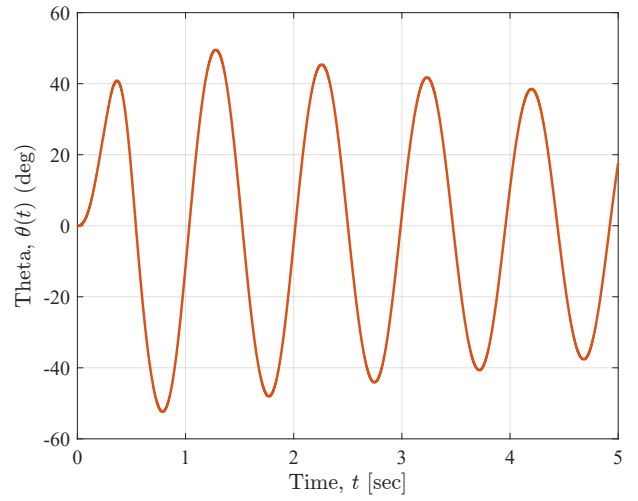


Figure 3.7: Tracking Closed-Loop with time-optimal reference, for 2DOF scheme.

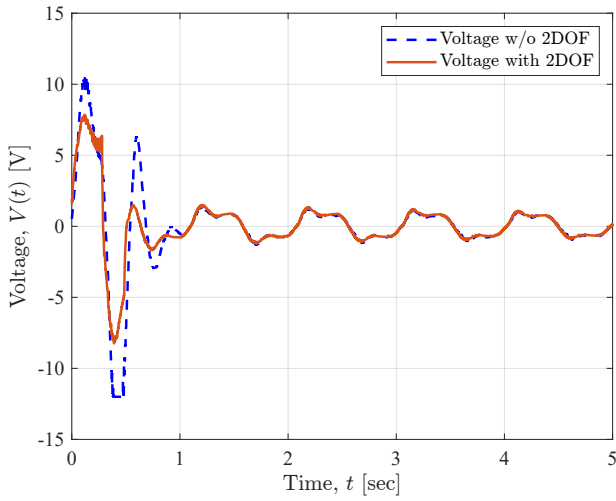
The motivation for using a 2DOF scheme is as follows: If $P_{\phi V}(s) = P_{\phi V}^{np}(s)$, and we feed-forward a control signal like in the scheme, we are guaranteed the optimal reference and voltage. But, because the systems are not identical, the mismatch between the systems is dealt with by the closed-loop in the 2DOF scheme, and we get the output and control signals to be similar to the desired signals.



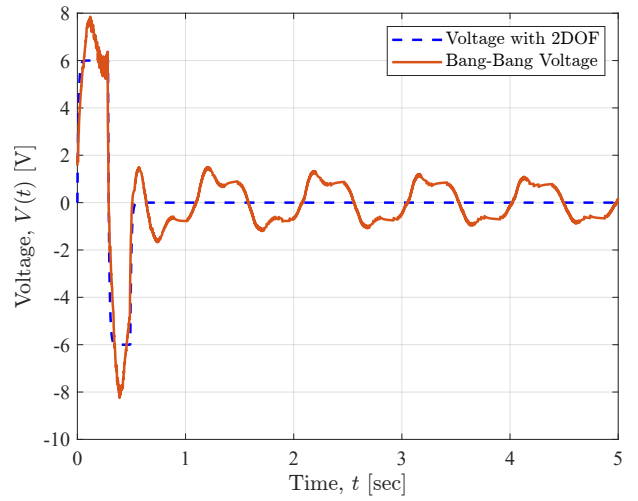
(a) Tracking performance: 1DOF vs. 2DOF.



(b) Response of the angle $\theta(t)$ for the 2DOF case.



(c) Comparison of $V(t)$ for 1DOF and 2DOF controllers.



(d) Comparison of $V(t)$ between Time-Optimal and 2DOF.

Figure 3.8: Performance analysis of Loosh controller tracking a Bang-Bang reference for $\phi_0 = 0$, $\phi_f = 60^\circ$, and $V_{max} = 6$. The sub-figures show tracking performance between 1DOF and 2DOF, voltage application, and θ response in the 2DOF scheme.

3.3.2 Compare Tracking of 1DOF and 2DOF Controller

In figure 3.8, we see that the tracking improved while using the 2DOF scheme, the voltage looks similar to the optimal one, as seen in 3.8d. However, the angle $\theta(t)$ oscillates in large amplitudes and as a result the arm doesn't fully rest when reaching the target angle.

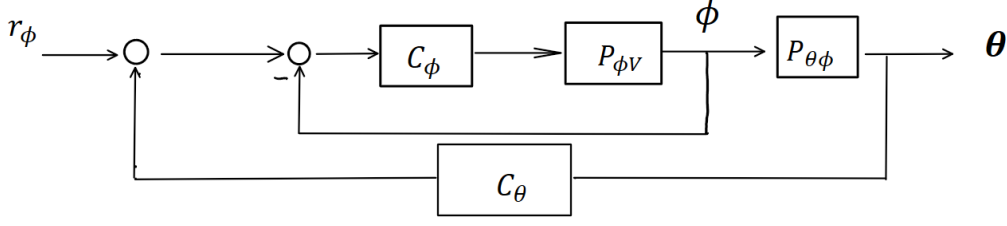


Figure 3.9: Scheme of tracking and damping closed-loop, for 1DOF tracking loop case.

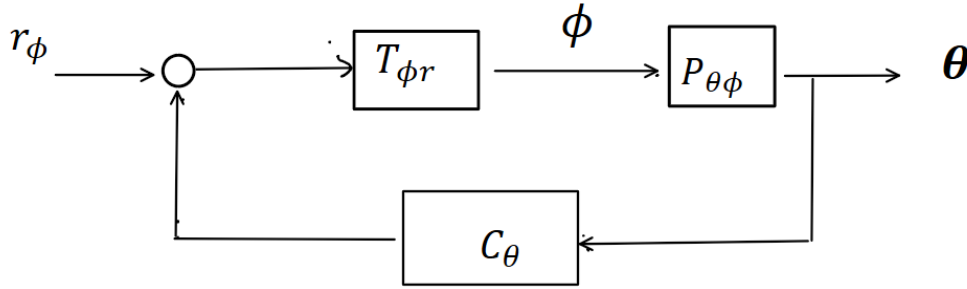


Figure 3.10: Simplified Scheme of tracking and damping closed-loop, for 1DOF tracking loop case.

3.4 Damping Controller Design with Loop-Shaping

In order to dampen the oscillations in $\phi(t)$, we designed a Loosh damping controller for the pendulum ($\theta(t)$). Damping the pendulum will also stop the oscillations in $\phi(t)$.

3.4.1 The Cascade Control Scheme of Damping + 2DOF Tracking

The systems only input is $V(t)$ and we wanted to both provide tracking and damping. Thus, we chose to design 2 control loops in cascade.

In figure 3.9 we can see the scheme, that adds another loop on top of the designed tracking closed-loop using a feedback controller $C_\theta(s)$.

3.4.2 Design of the Damping Controller

In the design of $C_\theta(s)$, we considered the Cascade-Control scheme of 1DOF Tracking + Damping. In figure 3.10 we see that for this case, $C_\theta(s)$ controls the inner closed-loop TF $T_{\phi r_\phi}(s) = \frac{P_{\phi V}(s)C_\phi(s)}{1+P_{\phi V}(s)C_\phi(s)}$ and $P_{\theta\phi}(s)$. The outer open-loop is $L_\theta(s) = -P_\theta(s)C_\theta(s)$, where $P_\theta(s) = T_{\phi r_\phi}(s)P_{\theta\phi}(s)$. Also, we denote the two crossover frequency's in $L_\theta(s)$ as $\omega_{c,1}$ and $\omega_{c,2}$.

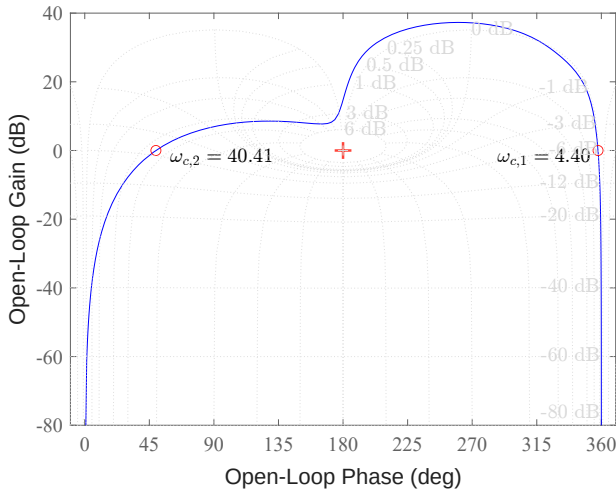
We require the following from the damping closed-loop:

1. closed-loop stability

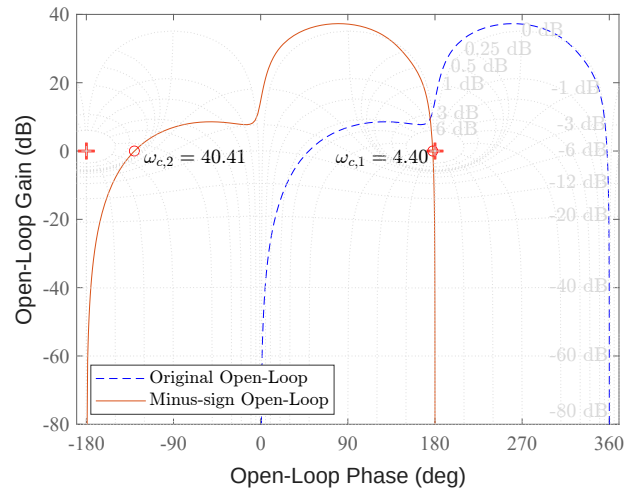
2. Adequate stability margins

3. We modify the controller and observe the change in the crossover frequency's $\omega_{c,1}, \omega_{c,2}$

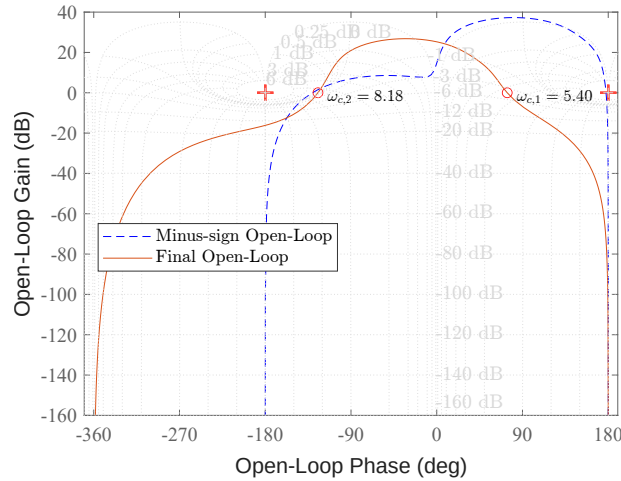
In figure 3.11, we can see the procedure of designing the damping controller. In the start, in figure 3.11a, the closed-loop isn't stable. For this reason, in 3.11b we chose the controller $C_\theta(s) = -1$ and we got a stable closed-loop, but with small PM near $\omega_{c,1}$. Finally, we added a 2nd-order LPF on the first crossover frequency $\omega_{c,1}$ to both increase the PM near $\omega_{c,1}$ and get the whole open-loop closer to the critical point near -180° . In figure 3.11c We see the final open-loop.



(a) Damping open-loop $L_\theta(s)$ with unity controller, meaning $C_\theta(s) = 1$.



(b) Compare damping open-loop before and after a minus sign.



(c) Compare damping open-loop after minus sign and final.

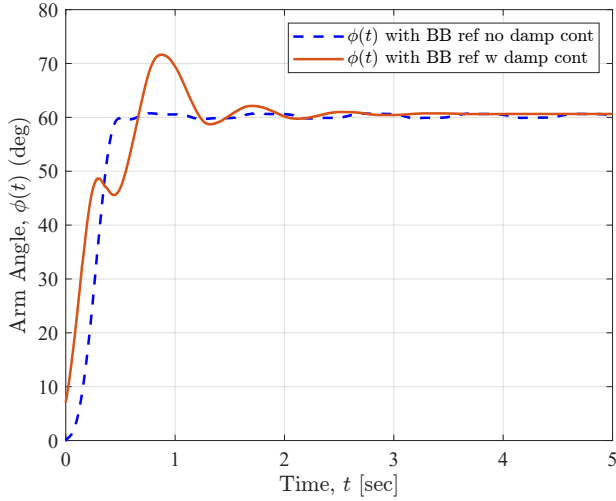
Figure 3.11: Damping controller design procedure, from original to final controller, including the crossover frequencies.

The resulting controller:

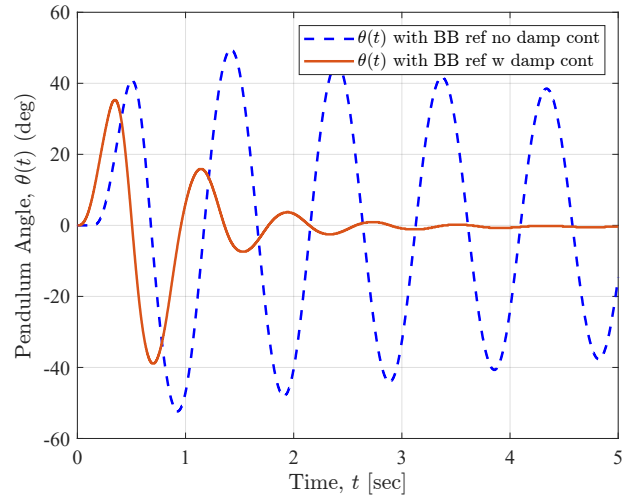
$$C_{\theta}(s) = \frac{-19.36}{s^2 + 8.8s + 19.36} \quad (3.3)$$

Closed loop stability is ensured through the Nichols diagram, and with the LPF in the controller we ensured desired PM and GM and got satisfactory crossover frequency's.

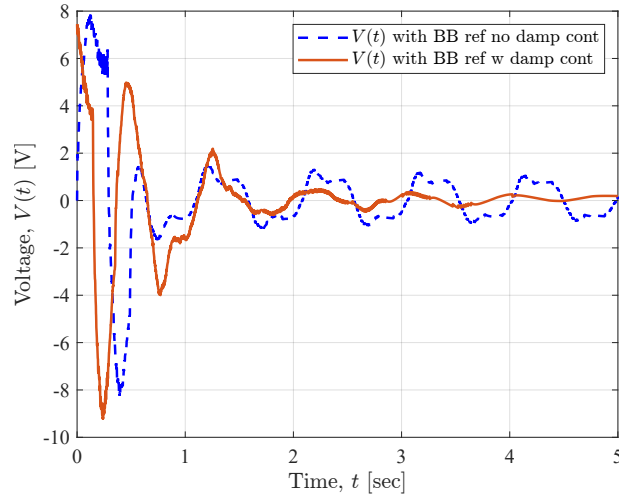
3.4.3 Tracking performance after adding the damping controller



(a) Response of the angle $\phi(t)$ for the with/without damping controller.



(b) Response of the angle $\theta(t)$ for the with/without damping controller.



(c) Comparison of $V(t)$ for the with/without damping controller.

Figure 3.12: Comparing output angles $\phi(t)$, $\theta(t)$ and the applied voltage $V(t)$ for input BB reference $\phi_f = 60^\circ$, for with and without damping controller.

In figure 3.12b, we see that $\theta(t)$ fades to zero after about 3[sec]. However, tracking worsened significantly. This is expected, as adding the dampening loop disturbs the reference signal.

3.5 Damping Controller Design with LQG

3.5.1 LQG problem and it's solution

Consider a stricly-proper SISO system $P(s)$ with it's state-space matrices $\begin{pmatrix} A & B \\ C & D=0 \end{pmatrix}$:

$$P: \begin{cases} \dot{x}(t) = Ax(t) + Bu(t) + Bd(t) \\ y(t) = Cx(t) + n(t) \end{cases}, \quad x(0) = 0 \quad (3.4)$$

Where $d(t), n(t)$ are white noise signals with intensities σ_d, σ_n , respectively. We want to find a stabilizing control signal $u(t)$, which will be a function of $y(t)$, that minimizes the following cost function:

$$J = \int_{t=0}^{\infty} \begin{pmatrix} x^T & u^T \end{pmatrix} \begin{pmatrix} Q & S \\ S^T & R \end{pmatrix} \begin{pmatrix} x \\ u \end{pmatrix} dt$$

It can be proven that the optimal solution for this problem is an observer-based state-feedback controller, where the SF is based on the LQR gains and the Luenberger estimator based on the Kalman-Filter gains (LQR+Kalman-Filter). Meaning, the solution is an estimator-based state feedback:

$$\begin{cases} \hat{\dot{x}} = A\hat{x} + Bu + L(y - C\hat{x}) \\ u = -K\hat{x} \end{cases} \Rightarrow \begin{cases} \hat{\dot{x}} = (A - BK - LC)\hat{x} + Ly \\ u = -K\hat{x} \end{cases} \quad (3.5)$$

Where the SF gain K is obtained from: $K = R^{-1}(S^T + B^T P)$, with P from the corresponding solution to the CARE problem of LQR, and the estimator gain L is obtained from: $L = \frac{1}{\sigma_n} Q C^T$, with L from the solution to the CARE problem of LQE.

3.5.2 Controller design procedure using an LQG solution

When giving the LQG solver any SISO process, we can get a stabilizing SISO controller as shown in 3.5. The process in which we will obtain an LQG-based $C(s)$, for process $P(s)$, is as follows:

1. Magnitude Shaping: We first introduce a shaper (or weight) in the form of the TF $W(s)$, which is designed to shape the open-loop's gain $L(s) = W(s)P(s) \equiv P_a(s)$ in a desired manner and add a necessary part into the controller.
2. Design an LQG controller for the augmented process $P_a(s)$: The augmented process is defined as follows:

$$P_a: \begin{cases} \dot{x}_a(t) = A_a x_a(t) + B_a u_a(t) + B_d a(t) \\ y(t) = C x_a(t) + n(t) \end{cases}, \quad x_a(0) = 0 \quad (3.6)$$

LQR control is designed with the following cost function weights:

$$J = \int_{t=0}^{\infty} (y^2 + u_a^2) dt = \int_{t=0}^{\infty} \begin{pmatrix} x_a^T & u_a^T \end{pmatrix} \begin{pmatrix} C_a^T C_a & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_a \\ u_a \end{pmatrix} dt \quad (3.7)$$

That is, equal weighting for the output and control effort in the augmented system. We will define unit-intensities for both white-noise signals.

- Thus, we obtain a controller $C_a(s)$ that ensured closed-loop stability for the process $P_a(s)$.

This approach provides a stable CL, but does not ensure a stable controller.

3.5.3 Design of damping controller based on LQG

We designed 3 different controller for the damping loop, using 3 different weights: $W(s) = \{0.1, 1, 10\}$. The controllers we obtained are stable. Therefore, we can check the stability in Nichols using the simplified stability criteria.

In figure 3.13, we see the different open-loops for each choice of the weight. In figure 3.13a, we see just a shift in amplitude when multiplying by $W(s) = \text{const.}$, but in figure 3.13b we see that when implementing the LQG solution on the augmented process, we get different open-loops for the damping CL.

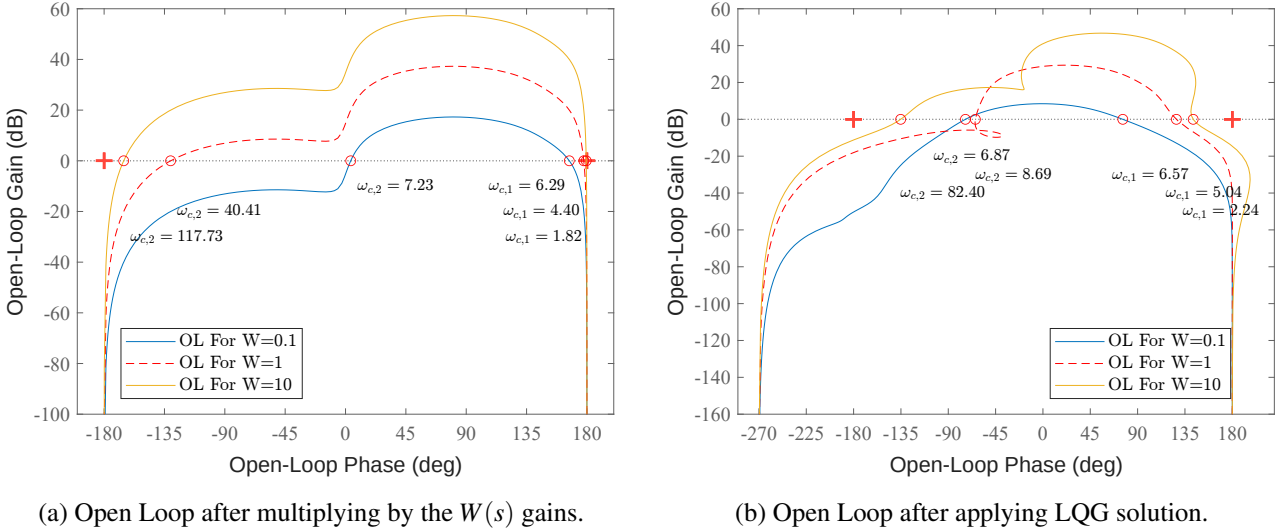


Figure 3.13: Design of Damping Controller $C_\theta(s)$ using LQG solution. On 3.13a is the OL for $C_\theta(s) = W(s)$ for the different weight. And on 3.13b is the OL after implementing a controller based on the LQG solution, for each case of $W(s)$.

The controller we got for $W = 1$:

$$C_\theta(s) = \frac{-24.13s^4 - 1003s^3 - 1.524 \times 10^4 s^2 - 1.808 \times 10^5 s - 4.087 \times 10^5}{s^5 + 94.25s^4 + 4628s^3 + 6.947 \times 10^4 s^2 + 3.021 \times 10^5 s + 7.029 \times 10^5} \quad (3.8)$$

3.5.4 Damping controller based on Loosh Vs. LQG, and LQG based on different weights

In figure 3.14, we see that dampening improves when using an LQG based $C_\theta(s)$ with $W = 1$, where in the LQG case, we went back to 1DOF tracking. We see that the required control signal is lower.

In figure 3.15, we see that the best performing damping controller is for $W = 1$.

From this point onward, we will use the damping controller in 3.8.

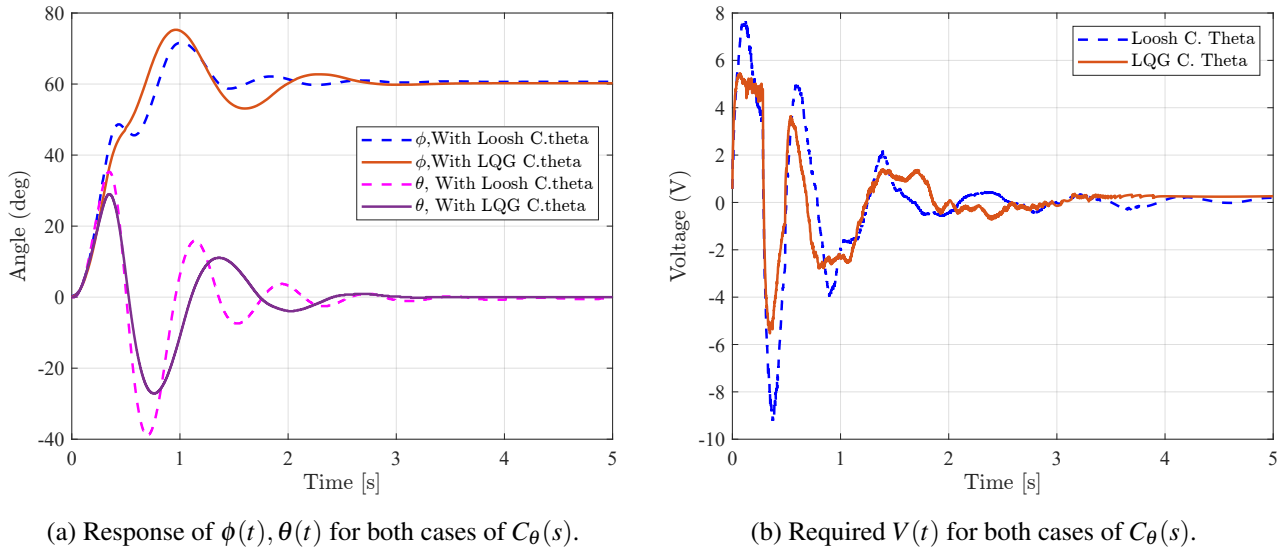


Figure 3.14: Comparing output angles $\phi(t), \theta(t)$ and the applied voltage $V(t)$ for input BB reference $\phi_f = 60^\circ$, for damping controller $C_\theta(s)$ based on Loop-Shaping Vs. LQG. for $W = 1$

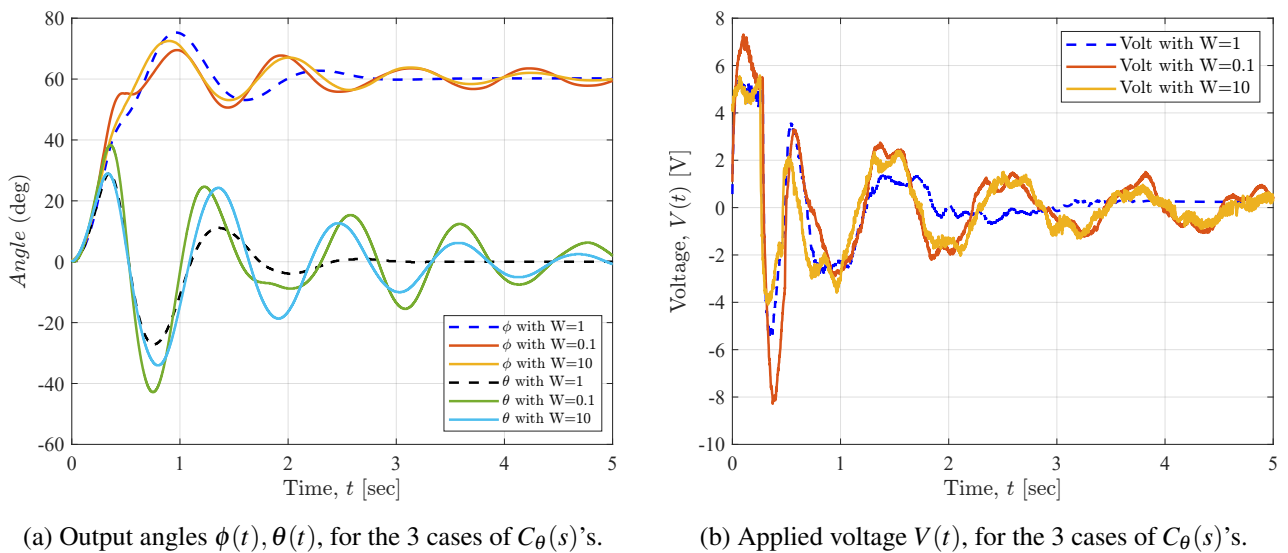


Figure 3.15: Comparing output angles $\phi(t), \theta(t)$ and the applied voltage $V(t)$ for input BB reference $\phi_f = 60^\circ$, for damping controller $C_\theta(s)$ based on LQG for different weight W 's chosen from $W(s) = \{0.1, 1, 10\}$.

Chapter 4

Intermittent Control Implementation

4.1 Motivation

Sometimes, the actuator and the sensor are not placed near each other, as shown in figure 4.1, and the only way for a near-analog control is to send large amounts of data through the net. Relying on the net has 2 drawbacks: It is not very reliable and can be expensive. By designing the system with intermittent control in mind, we can significantly reduce the sample rate while still maintaining good system performance.

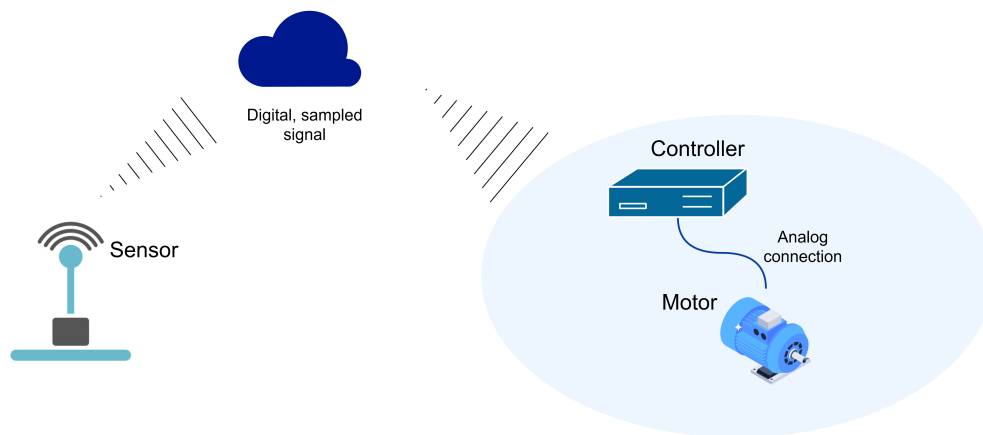


Figure 4.1: An example scheme for a net-based control system

4.2 The basic idea

The purpose of this strategy is to take an existing and stabilizing controller with adequate performance and introduce intermittency in the communication link between the sensors and the controller. In digital control, a holder like ZOH is being used, and the system works in an open loop between the sampling times, so when a low sampling frequency is used in the scheme, the control design may work poorly or not at all.

To improve the performance of the system, we chose to use a special hold block with the knowledge of the system dynamics and the reference input. This block allows the controller to accurately estimate the required control signal without frequent sampling if disturbances are not present.

4.3 Implementing an HOLD block with constant sampling time

Let us first compare the scheme of the previous controller and the decomposed one. We want to decompose $C_\theta(s)$ from figure 3.10 into its observer and SF from eq. 3.5. $P_\theta(s)$'s state-space representation will be:

$$P_\theta : \begin{cases} \dot{x}_\theta(t) = A_\theta x_\theta(t) + B_\theta u_\theta(t) \\ \theta(t) = C_\theta x_\theta(t) \end{cases} \quad (4.1)$$

Observers need to obtain the entire input of the system $P_\theta(s)$, which includes the reference, to truly estimate the state vector. Thus, we need to formulate the inner loop's dynamics from inputs to output and use it in the observer's equation.

Firstly, let's define the relevant systems in state-space. The inner closed-loop of tracking $T_{\phi r_\phi}(s)$ and the process between the angles $P_{\theta\phi}(s)$:

$$T_{\phi r_\phi} : \begin{cases} \dot{x}_1(t) = A_1 x_1(t) + B_1 u_\theta(t) \\ \phi(t) = C_1 x_1(t) \end{cases}, \quad P_{\theta\phi} : \begin{cases} \dot{x}_2(t) = A_2 x_2(t) + B_2 \phi(t) \\ \theta(t) = C_2 x_2(t) + D_2 \phi(t) \end{cases} \quad (4.2)$$

And for the process of a new LPF used in this scheme: $P_{LPF}(s) = \frac{1}{\tau s + 1}$ where $\tau = 0.05$:

$$P_{LPF} : \begin{cases} \dot{x}_L(t) = A_L x_L(t) + B_L u_1(t) \\ u_F(t) = C_L x_L(t) \end{cases} \quad (4.3)$$

Now, the representation in state-space of the entire system, from $(u_1(t), r_\phi(t))^T$ to $\theta(t)$, with $x_w = (x_L, x_1, x_2)^T$:

$$\begin{cases} \dot{x}_w(t) = A_w x_w(t) + B_w u_1(t) + B_{r_\phi} r_\phi(t) \\ \theta(t) = C_t x_w(t) \end{cases} \quad (4.4)$$

In addition $\phi(t) = C_w x_w(t)$. Where:

$$A_w = \begin{pmatrix} A_L & 0 & 0 \\ B_1 C_L & A_1 & 0 \\ 0 & B_2 C_1 & A_2 \end{pmatrix}, \quad B_w = \begin{pmatrix} 0 \\ B_1 \\ 0 \end{pmatrix}, \quad B_{r_\phi} = \begin{pmatrix} B_L \\ 0 \\ 0 \end{pmatrix} \quad (4.5)$$

And:

$$C_t = \begin{pmatrix} 0 & D_2 C_1 & C_2 \end{pmatrix}, \quad C_w = \begin{pmatrix} 0 & C_1 & 0 \end{pmatrix} \quad (4.6)$$

From eq .4.4 we see that the equation of the observer should be:

$$\dot{\hat{x}}_w(t) = A_w \hat{x}_w(t) + B_w u_1(t) + B_{r_\phi} r_\phi(t) + L(\theta(t) - C_t \hat{x}_w(t)) \quad (4.7)$$

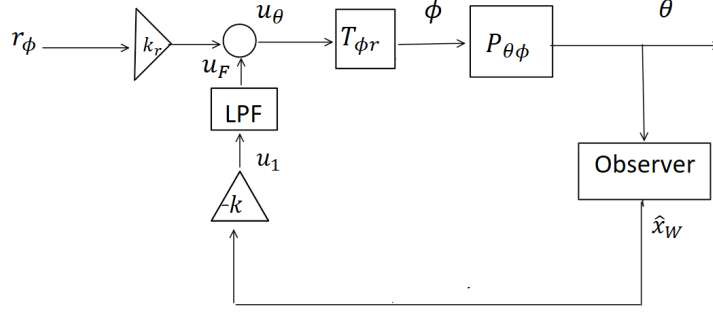


Figure 4.2: Control scheme after decomposing $C_\theta(s)$ into observer + state-feedback, with observer receiving the systems full input of $u_1(t)$ and $r_\phi(t)$, and adding a LPF.

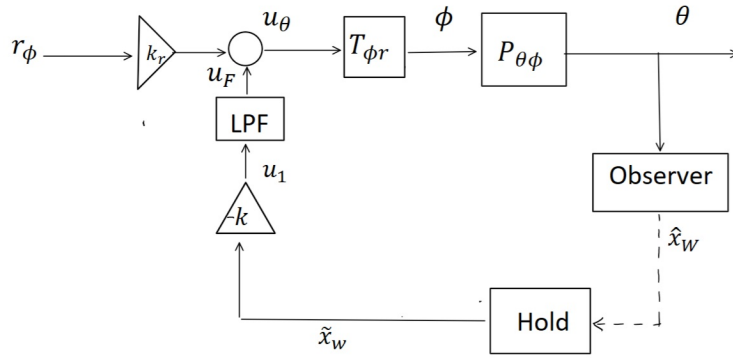


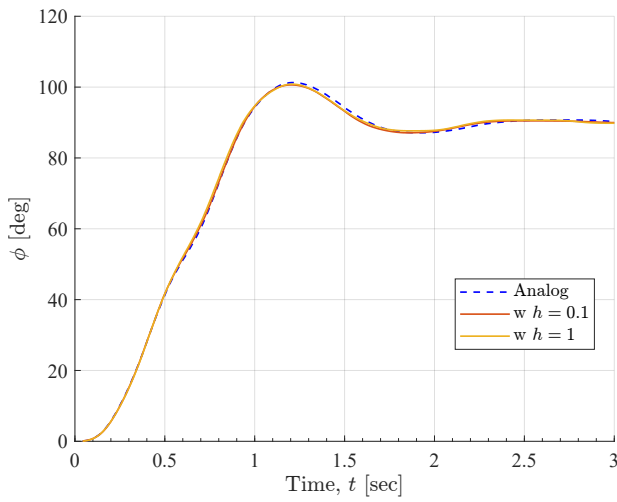
Figure 4.3: Control Scheme decomposed with holder.

Thus, we designed an observer+state-feedback damping controller for the whole system described in eq .4.4, including the LPF. We added a gain k_r to get zero steady-state error in the outer loop. The scheme we got is shown in figure 4.2. The scheme used for intermittent control of the pendulums damping is shown in figure 4.3. The dynamics of the HOLD block is chosen to be that of the inner loop:

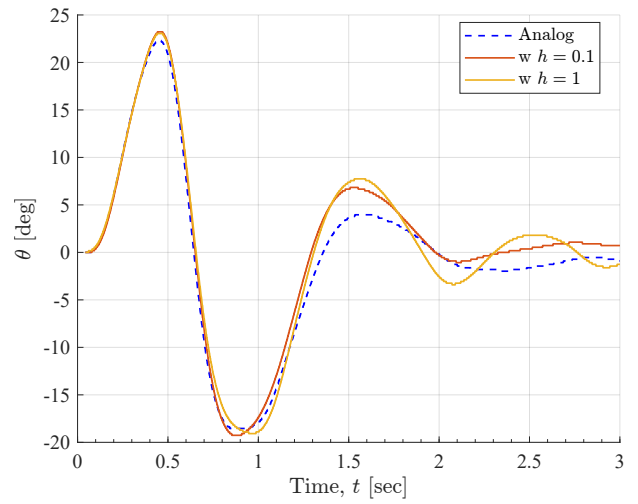
$$\begin{cases} \dot{x}_h(t) = A_w x_h(t) + B_w u_1(t) + B_{r_\phi} r_\phi(t) \\ x_h(ih) = \hat{x}_w(ih), \quad i \in \mathbb{N} \end{cases} \quad (4.8)$$

Where h is constant for now, but can be chosen to be dynamic.

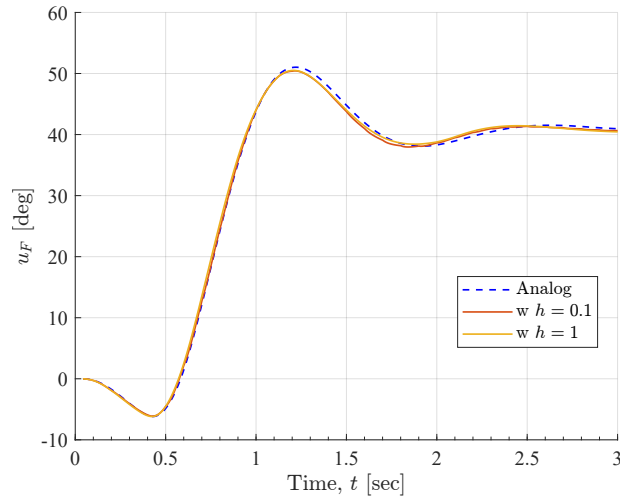
The results of the experiments are shown in figure 4.4. We can see that when the HOLD block has knowledge of the whole inner system input, including the reference, the system doesn't behave differently, even for large sampling times like $h = 1[\text{sec}]$. Meaning, giving the HOLD block this knowledge allowed the controller to get less data from the sensors side of the control scheme.



(a) comparing $\phi(t)$ for analog case with no HOLD block, and sampling times $h = 0.1, 1$.



(b) comparing $\theta(t)$ for analog case with no HOLD block, and sampling times $h = 0.1, 1$.



(c) comparing $u_F(t)$ for analog case with no HOLD block, and sampling times $h = 0.1, 1$.

Figure 4.4: tracking control, comparing ϕ, θ, u_F for analog case with no HOLD block, and for cases with sampling times $h = 0.1, 1$.

4.4 Event triggered control for dampening

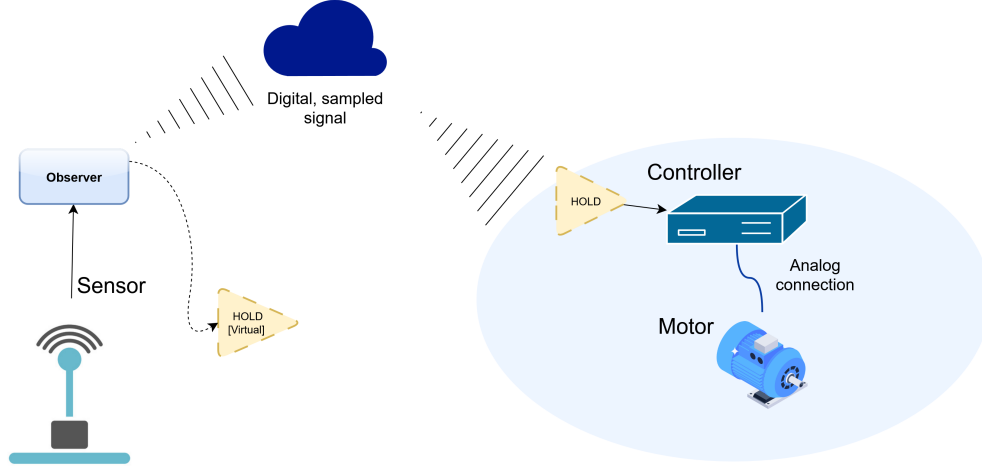


Figure 4.5: An example scheme for a control system with event triggered sampling.

In order to base the dampening on event-triggered control, we need to formulate the conditions for switching between fast and slow sampling times. We decided to implement a dynamic $h(t)$ using a piecewise-constant function approach. We have chosen the following criteria:

$$h(t) = \begin{cases} h_{\text{slow}} = 0.5[\text{sec}] & |\theta - \hat{\theta}| < 3^\circ \\ h_{\text{fast}} = 0.1[\text{sec}] & |\theta - \hat{\theta}| \geq 3^\circ \end{cases} \quad (4.9)$$

Here $\hat{\theta}$ is taken from the state vector of the HOLD block.

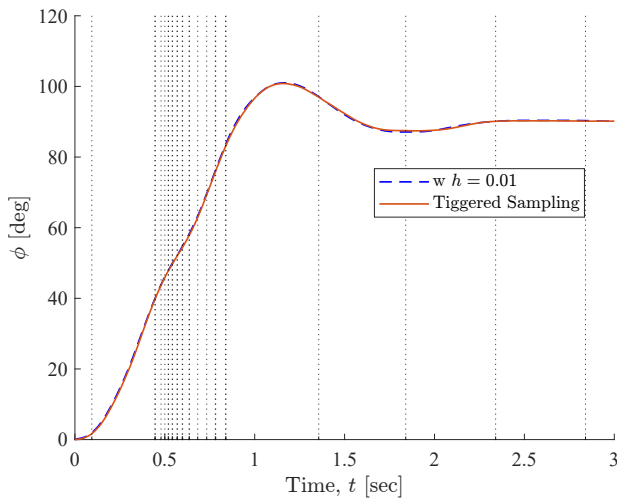
In practice, a virtual HOLD block is connected to the observer on the sensor side, using measurements and observer estimation of $\theta(t)$ to determine switching times, as shown in Fig. 4.5.

We calculated the average sampling time h_{avg} using the following formula:

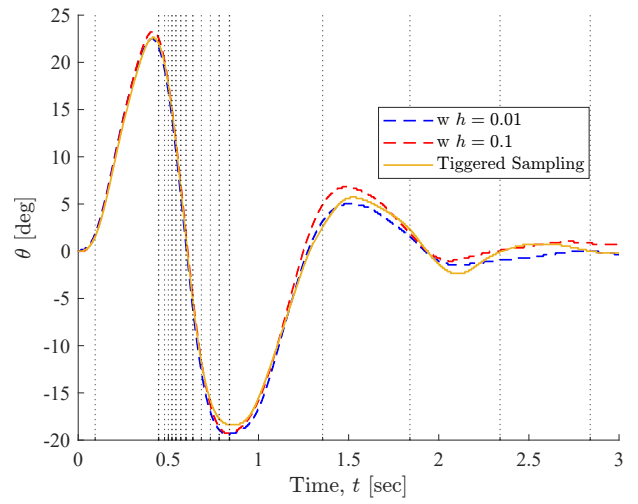
$$h_{\text{avg}} = \frac{\text{total time}}{\text{No. of rises of Pulse}}$$

In figure 4.6 the average sampling time is $h_{\text{avg}} = 0.126$. We see that the dynamical response is almost the same, even with the average sampling time obtained, and the response is a bit better than with $h = 0.1$.

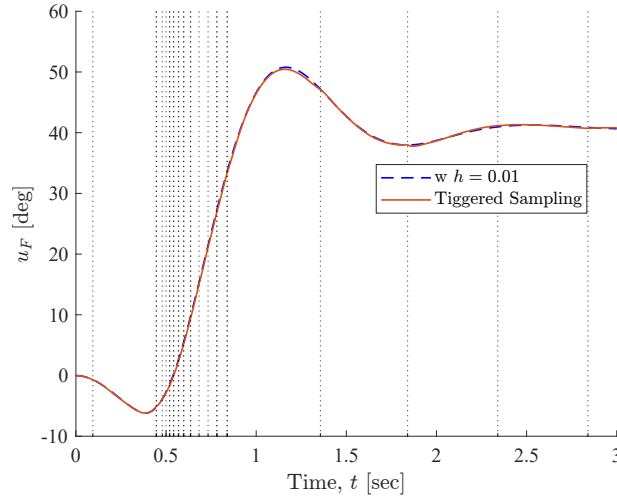
Note: The sampling was implemented by a pulse entering the integrator inside the Hold block, with 2 different frequencies. With every switch between the two, the pulse generator started with a brief "1" value. With the fact that the switching decision is "Analog" ($f_s = 1[\text{kHz}]$), the actual sampling time will be smaller than the "fast" one if the system jumps between sampling times. This can be seen when $\dot{\theta}$ is high in figure 4.6. One way to solve this issue is to set the fast sampling as the general sampling time of the system, or choose h_{fast} as a rate larger than the systems $1/f_s = 0.001$ and smaller than h_{slow} , while implementing a smearing mechanism that will let the system stay in the fast mode to prevent chattering.



(a) Comparing $\phi(t)$ for sampling time $h = 0.01$ and event triggered $h(t)$.



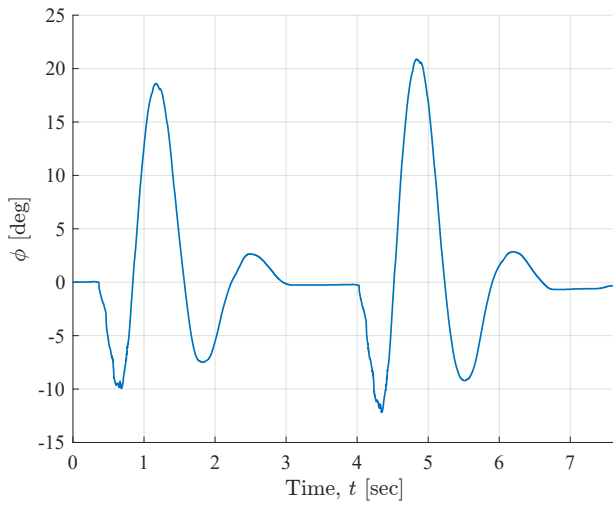
(b) Comparing $\theta(t)$ for sampling time $h = 0.1, 0.01$ and event triggered $h(t)$.



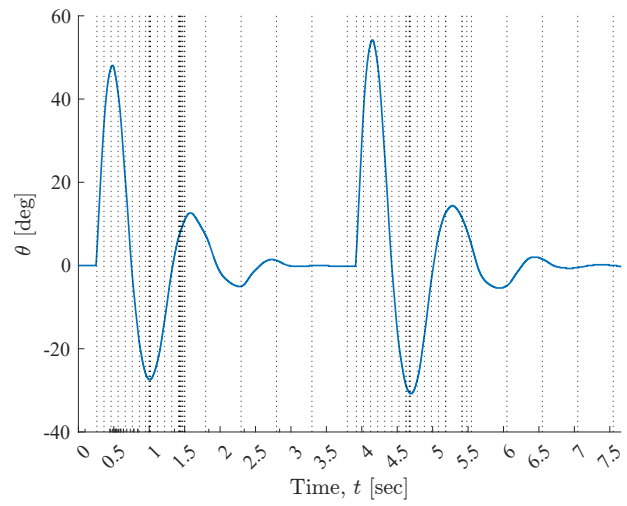
(c) Comparing $u_F(t)$ for sampling time $h = 0.01$ and event triggered $h(t)$.

Figure 4.6: Intermittent tracking control: comparison of $\phi(t)$, $\theta(t)$, and $u_F(t)$ for constant sampling time $h = 0.01$, and for a dynamic sampling time $h(t)$, with indicators of samples taken. Only for $\theta(t)$, we display the case with $h = 0.1$, as the differences between these three cases are most clearly visible in this signal.

In figure 4.7, when the system responds to external disturbances, in this case impact, while tracking $\phi \rightarrow 0$, the sampling rate increases more frequently than during tracking. These high-rate pulses occur immediately after each hit. The densely sampled region presents the same challenge observed in the previous figure.



(a) $\phi(t)$ response to disturbance for $h(t)$.



(b) $\theta(t)$ disturbance response for $h(t)$, with samples.

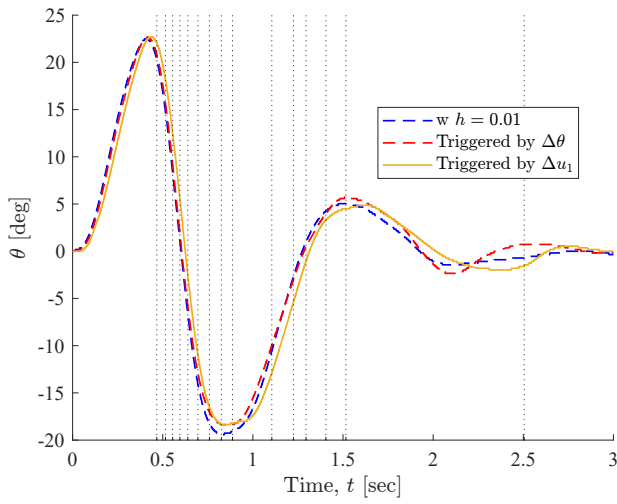
Figure 4.7: Intermittent tracking control, ϕ, θ response to disturbance for dynamic $h(t)$. Only for $\theta(t)$, we display the indicators of sampling times.

Because the system samples with h_{slow} , even though the error is almost zero, we decided on a rule of $h(t)$ that does not sample when not needed. When the oscillations of the pendulum have a low amplitude, the previous criterion of comparing $\theta(t)$ and $\hat{\theta}(t)$ prevents the controller from fully damping the pendulum. This led us to choose the following criteria, based on the control signal $u_1(t)$, which is a linear combination of all states, not just $\theta(t)$:

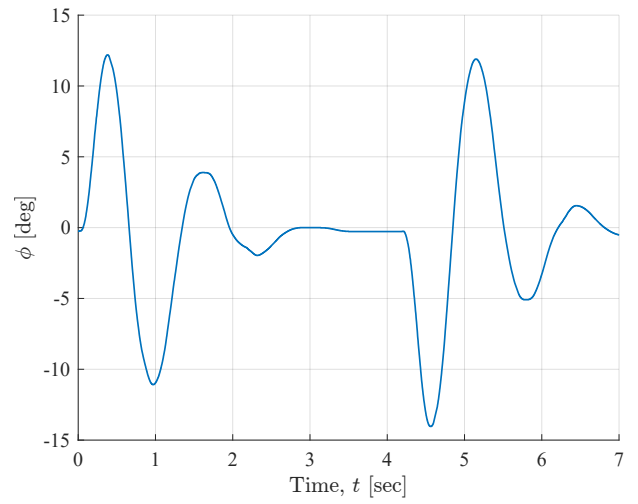
$$h(t) = \begin{cases} \text{no sampling} & |u_1 - u_{1,\text{obs}}| < 0.012 \\ h_{\text{fast}} = 0.01 & |u_1 - u_{1,\text{obs}}| \geq 0.012 \end{cases} \quad (4.10)$$

Where u_1 is taken from the HOLD block and $u_{1,\text{obs}}$ from the observer.

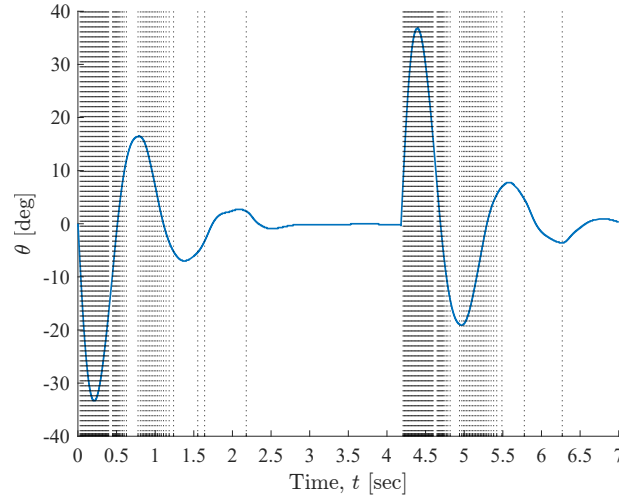
In figure 4.8, the average sampling time is $h_{\text{avg}} = 0.2$ and we see that the behavior of $\theta(t)$ under disturbance and tracking is similar between the cases of sampling triggered by the difference in θ or u_1 . However, sampling rate decreases when the system is almost at rest. This can reduce data transfer.



(a) Comparing $\theta(t)$ for sampling time $h = 0.01$ and dynamic $h(t)$ based on $\theta(t)$ and $u_1(t)$ difference.



(b) $\phi(t)$ for dynamic sampling time $h(t)$ based on $u_1(t)$ difference.



(c) $\theta(t)$ for dynamic sampling time $h(t)$ based on $u_1(t)$ difference.

Figure 4.8: Intermittent tracking control: comparison of $\phi(t)$ between the case of $h = 0.01$ and for a dynamic sampling time $h(t)$ based on $\theta(t)$ or $u_1(t)$ difference, with indicators of samples taken. And the response of $\phi(t)$, $\theta(t)$ for impact disturbance, for $h(t)$ based on $u_1(t)$ difference, with indicators of samples taken only on $\theta(t)$ plot.

Chapter 5

Bonus: Stabilizing the pendulum in upper equilibrium

From the TF's $P_{\theta\phi}(s), P_{\phi V}(s)$ in equations 1.7,1.11 for the bottom eq. and 1.14,1.16 for the top, we see that we can get $P_{\theta\delta\phi}(s)$ straight from $P_{\theta\phi}(s)$ and $P_{\phi V}^t(s)$ from $P_{\phi V}(s)$ using the assumption that: $(J_a + m_p r^2)m_p l_c g \gg f_a f_p + \frac{(KN_g)^2}{R} f_p$. Using this we get:

$$\begin{cases} P_{\phi V}^t(s) = \frac{3.08s^2 + 1.219s - 137.1}{s(s^3 + 1.451s^2 - 49.01s - 50.72)} \\ P_{\theta\phi}(s) = \frac{-1.261s^2}{s^2 + 0.115s - 45.3} \end{cases} \quad (5.1)$$

The scheme implemented for this case is shown in figure 5.1. We desired to stabilize the system just like the bottom case: first stabilize $P_{\phi V}^t(s)$ using $C_{\phi}^t(s)$ in the inner loop, and then stabilize $P_{\theta\delta\phi}(s)$ + the inner loop using $C_{\theta}^t(s)$ in the outer loop. But when observing the poles and zeros of $P_{\phi V}^t(s)$ we see:

$$P_{\phi V}^t(s) = \frac{3.08(s + 6.873)(s - 6.477)}{s(s - 6.822)(s + 7.248)(s + 1.026)}$$

Meaning, this process has close RHP zero and pole and thus can be controlled only with small stability margins. This corresponds to the control task of stabilizing $\phi(t)$ and $\theta(t)$, using only the measurement of $\phi(t)$, like trying to stabilize an inverted pendulum with a blindfold.

In this case, we decided on the following design procedure:

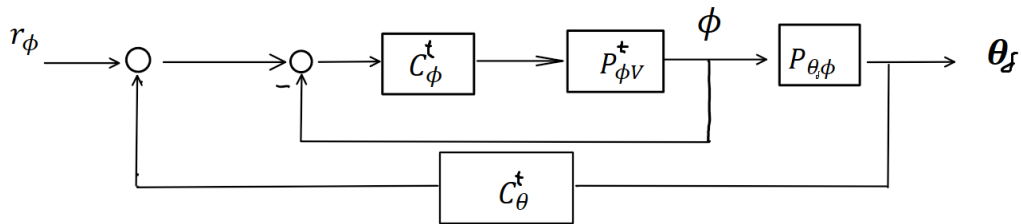


Figure 5.1: Scheme of tracking $\phi(t)$ and stabilizing $\theta(t)$ closed-loop.

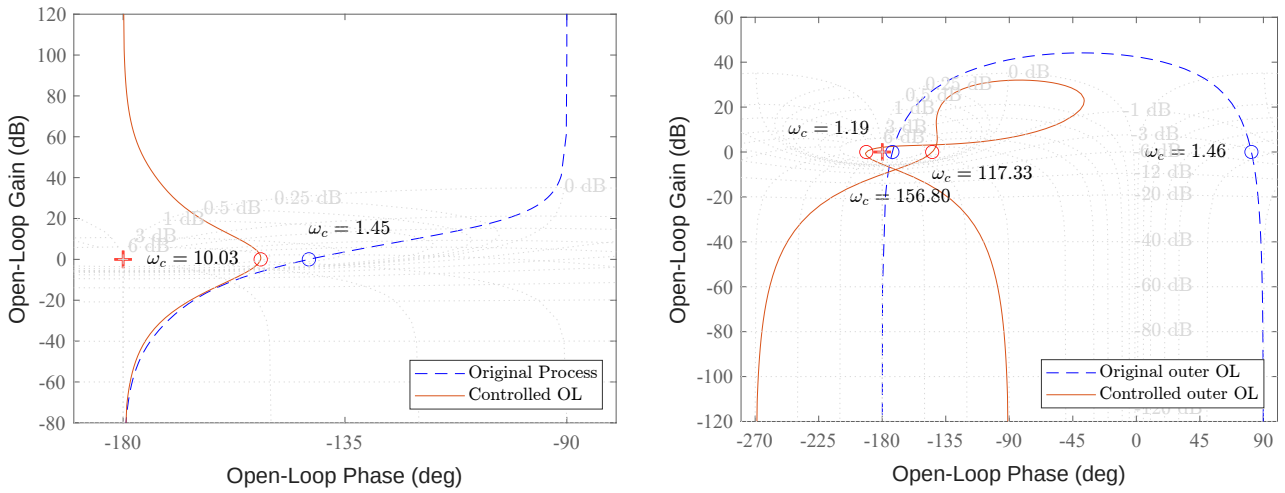
1. Design $C_\phi^t(s)$ for the simplified process: $P_{\phi V, \text{simp}}^t(s) = \frac{3.08(s+6.873)}{s(s+7.248)(s+1.026)}$ ensuring stability, zero steady-state error for static reference and disturbance and adequate stability margins.
2. Design $C_\theta^t(s)$ for the **real** outer loop and stabilizing it, using an LQG based controller.

In the lab, we modified the above processes in order to stabilize the real system. We worked with the following TF's:

$$\begin{cases} P_{\phi V}^t(s) = \frac{2.871(s+6.827)(s-6.712)}{s(s-7.156)(s+7.471)(s+1.082)} \\ P_{\theta \phi}(s) = \frac{-1.261s^2}{s^2+0.115s-45.82} \\ P_{\phi V, \text{simp}}^t(s) = \frac{2.871(s+6.827)}{s(s+7.471)(s+1.082)} \end{cases} \quad (5.2)$$

Firstly, we stabilized the simplified system $P_{\phi V, \text{simp}}^t(s)$ in a similar vain to stabilizing the bottom equilibrium process between voltage and arm angle. We used a controller $C_\phi^t(s)$ that includes gain for increasing ω_c , lag in low freq. to include an integrator and a lead to increase PM, as seen in figure 5.2a.

We made the controller $C_\theta^t(s)$ using a LPF weight $W(s) = \frac{\omega_m}{s+\omega_m}$. While tuning the inner controller and ω_m , we reached the outer open loop seen in figure 5.2b that also stabilizes the real system.

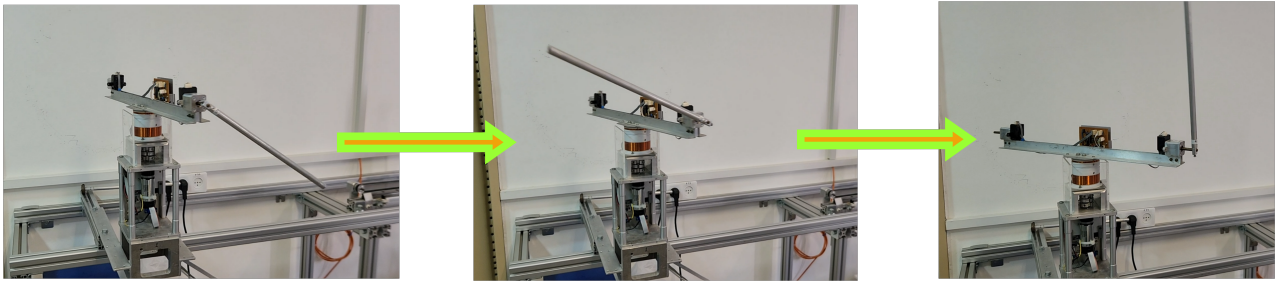


(a) Comparing inner open-loop between original $P_{\phi V, \text{simp}}^t(s)$ and controlled open loop $P_{\phi V, \text{simp}}^t(s)C_\phi^t(s)$. (b) Comparing outer open-loop before and after adding $C_\theta^t(s)$.

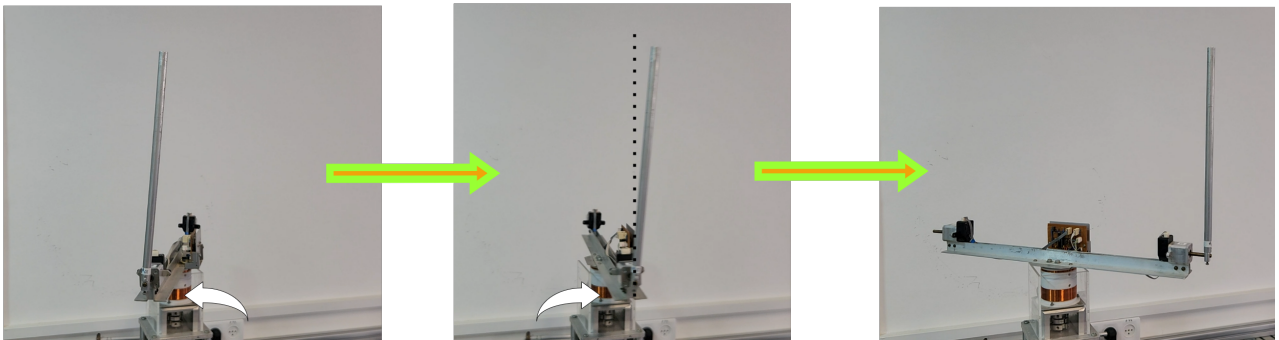
Figure 5.2: Process of cascade control open-loop design for the pendulum in top equilibrium.

We saw that in the top equilibrium, the stability margins are lower than for the bottom one. As a result, system uncertainties effect the closed loop significantly. This required us to repetitively modify the controller, in order to obtain one that stabilizes both the model and the real process.

The following images in 5.3 show the erecting of the pendulum and tracking process. Erecting was done by applying a sinusoidal voltage signal with tuned freq. that matches the resonance of the process $P_{\theta V}^t(s)$.



(a) Erecting process and switching to upper stabilizing controller.



(b) Arm angle reference tracking. The system undershoots in order to track the angle.

Figure 5.3: Images showing the process of erecting the pendulum to upper eq., stabilizing it around the top eq. and zero arm angle, and then tracking the reference arm angle.

Chapter 6

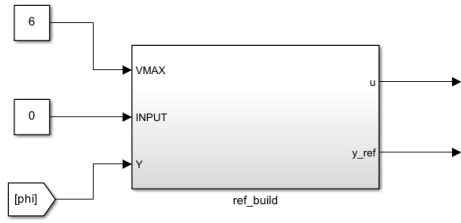
Appendix : Specific block diagrams used

In this part, we will briefly go through block diagrams that are key to our implementation of the event-triggered control scheme.

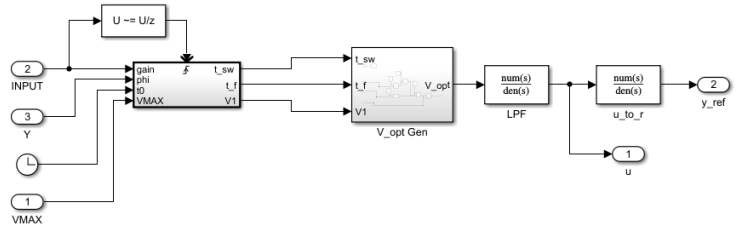
6.1 Time-Optimal reference Generator

As seen in figure 3.4, the generator receives the desired final angle ϕ_f , calculates the required control signal and time-Optimal reference $r_\phi(t)$, assuming simplified process $P_{\phi V}^{Virtual}(s)$.

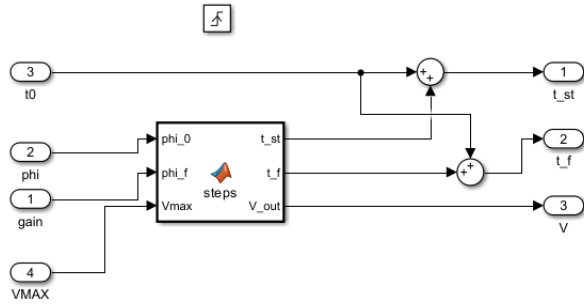
The block diagram used is seen in figure 6.1. The reference generator is inside the ref-build block, as seen in fig. 6.1a, it receives $V_{\max}$, ϕ_f and current $\phi(t)$ as Vmax, INPUT and Y. Fig. 6.1b shows the inner structure, including a Triggered-Subsystem block that takes the inputted signals, with the current time, just when $\phi(t)$ changes (mark change time t_0), and calculates the required switching times (with t_0) and initial voltage, inside of the Triggered-Subsystem in fig. 6.1c. The following Vopt Gen block in fig 6.1d calculates $V_{\text{opt}}(t)$ for the current time. This voltage is filtered through a LPF and passes $P_{\phi V}^{Virtual}(s)$ to get both the optimal voltage and reference.



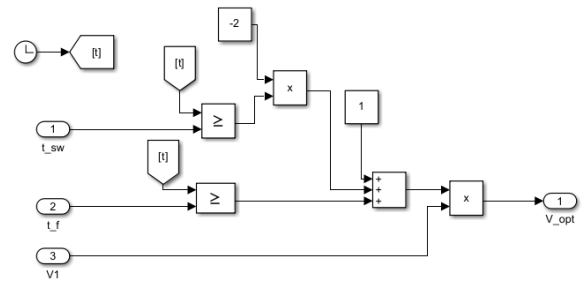
(a) The block diagram for generating the reference.



(b) Inner Structure of the subsystem ref-build.



(c) Triggered-Subsystem block that takes the inputted signals to calculate the required switching times and initial voltage.



(d) Vopt Gen block calculates $V_{opt}(t)$ for current time.

Figure 6.1: This block diagram generates a reference and required control signal using the current $\phi(t)$, target ϕ_f , and max voltage V_{max} (fig. 6.1a). The inner structures used include: a ref-build subsystem with a Triggered Subsystem for calculating switching times and initial voltage for obtaining desired ϕ_f (figures 6.1b, 6.1c), an optimal voltage block for calculating the required voltage signal (fig. 6.1d) and two transfer functions for filtering and reference output.

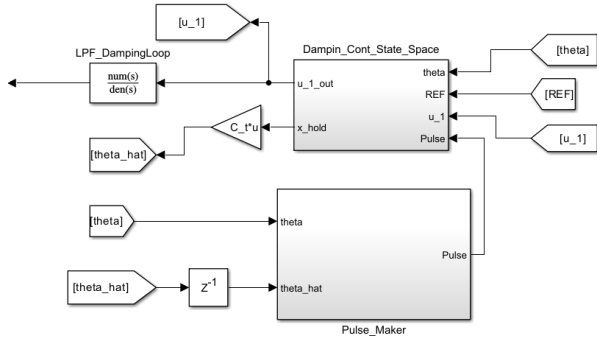
6.2 Event-triggering mechanism and Resetting emulator for the HOLD block

The control scheme used in event triggered control is as seen in fig. 4.3. We will focus here on the implementation of event-triggered control in the outer loop, specifically for the law seen in eq. 4.9.

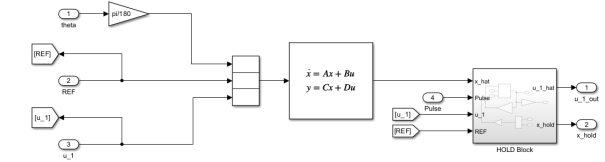
In figure 6.2a we see the outer loop from $\theta(t)$ to $u_F(t)$, in which the Damping Control subsystem, in fig. 6.2b, receives the measured $\theta(t)$, $r_\phi(t)$, $u_1(t)$, and the desired sampling pulse signal, and outputs $u_1(t)$ from the HOLD block and the observer's state $\hat{x}(t)$. $u_1(t)$ is summed with the reference after that to complete the loop and both $\theta(t)$ and $\hat{\theta}(t)$ from $\hat{x}(t)$ are used to determine the event-triggering in the Pulse-Maker block, in fig. 6.2d, which gives a pulse back to the Damping Control subsystem.

In addition, to initialize the state vector in the HOLD block dynamics, as desired in eq. 4.8, we need to make a state space system that can be initialized when we desire to take a sample. We needed to separate the state-space

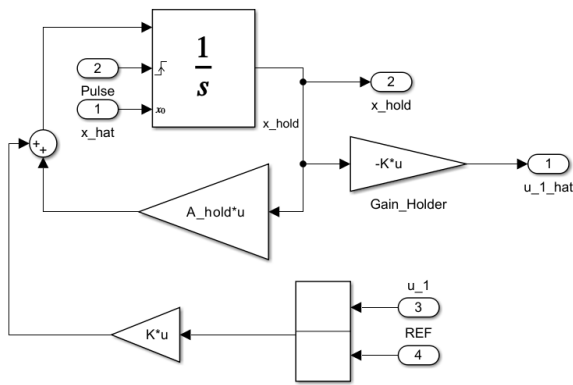
realization to integral and gain structure to implement the resetting of the integral using a pulse and IC from the observer, as seen in figure 6.2c.



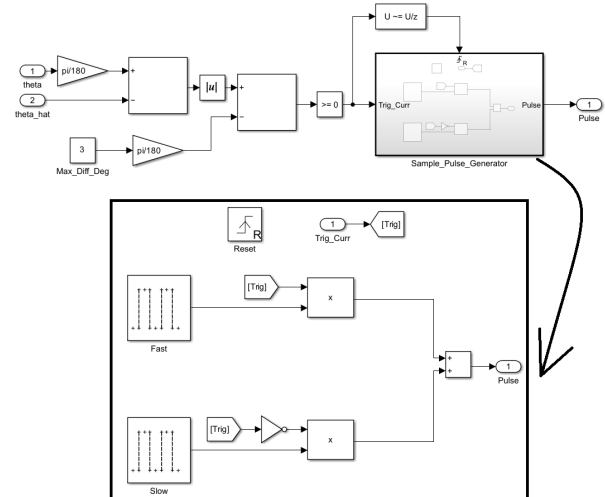
(a) The feedback block diagram for dampening the pendulum using event-triggered scheme.



(b) Damping-Control subsystem with observer that receives feedback θ , reference and control signal inputted into the inner loop, and outputs both $u_1(t)$ and $\hat{x}(t)$.



(c) Inner subsystem of the HOLD block, including reset emulator for the hold function.



(d) Pulse-Maker subsystem block diagram, specifically for the case of the event-triggered law in eq. 4.9.

Figure 6.2: The block diagram shows the feedback part of the control scheme, with the modifications to have event-triggered control for the law in eq. 4.9. It receives the feedback θ , reference, control signal u_1 and the desired sampling pulse. The Damping subsystem outputs $u_1(t)$ and $\hat{x}(t)$ and the Pulse-Maker block compared both $\theta(t)$ and $\hat{\theta}(t)$ to decide on the Pulse for the current time.

Summary

This project marked our first hands-on experience in designing a control system for a real physical setup. We began with system modeling and identification, then developed a basic unity-feedback tracking controller for the arm's angle. Over time, we refined the control law to enhance performance.

Working with a real system introduced challenges that were not always apparent during simulation. These occurred because of system uncertainties, model nonlinearities and mechanical changes that happens to the system during the course of the project. These issues highlighted the critical importance of both control robustness and accurate modeling when dealing with real-world systems.

Counterintuitively, when we implemented intermittency in the tracking and damping control law, we observed that it was possible to reduce the average sampling rate between the sensors and the controller while still improving overall system performance.

We would like to thank Prof. Leonid Mirkin and Ilya Shamis for their dedicated guidance and valuable support throughout the project.